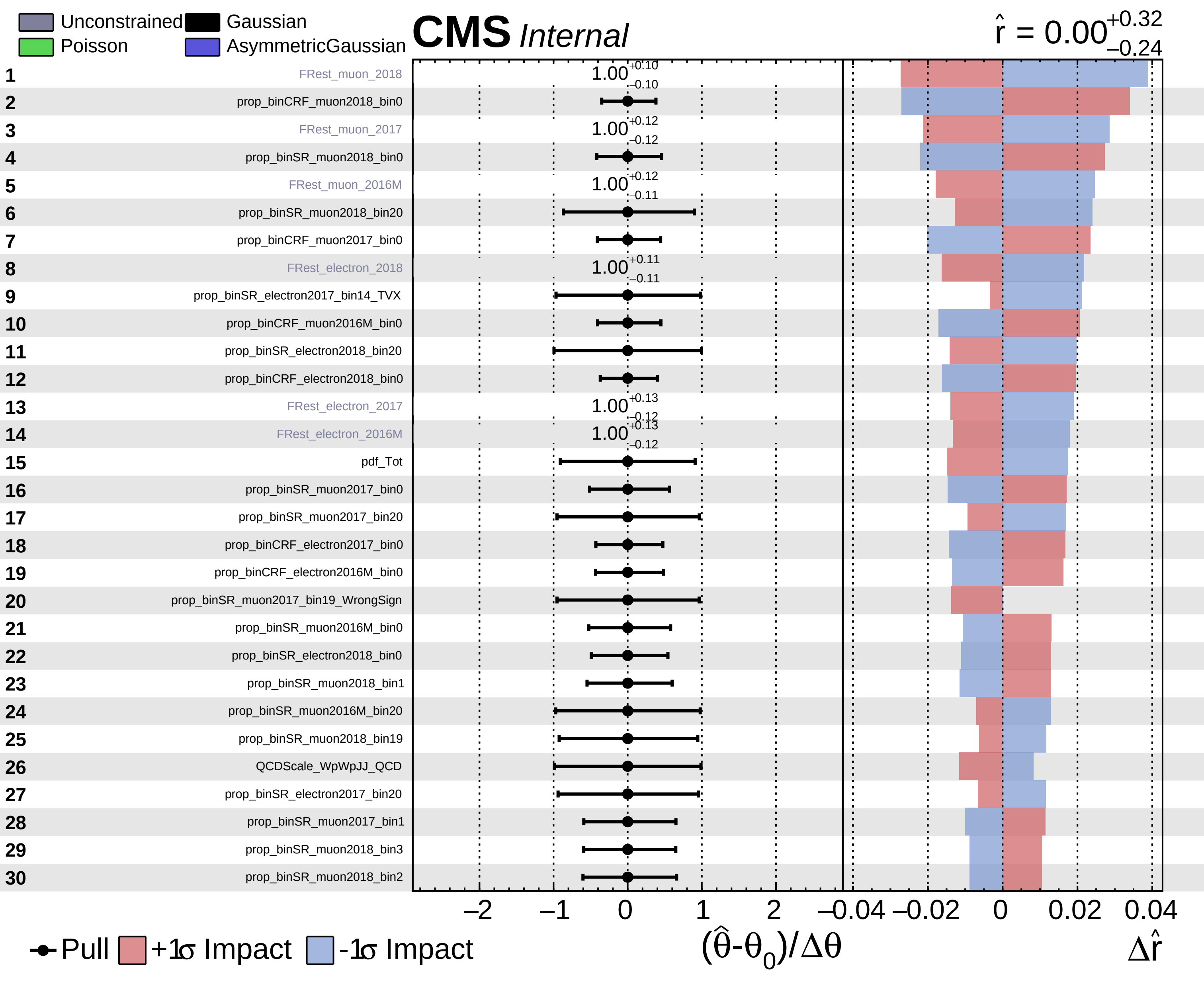
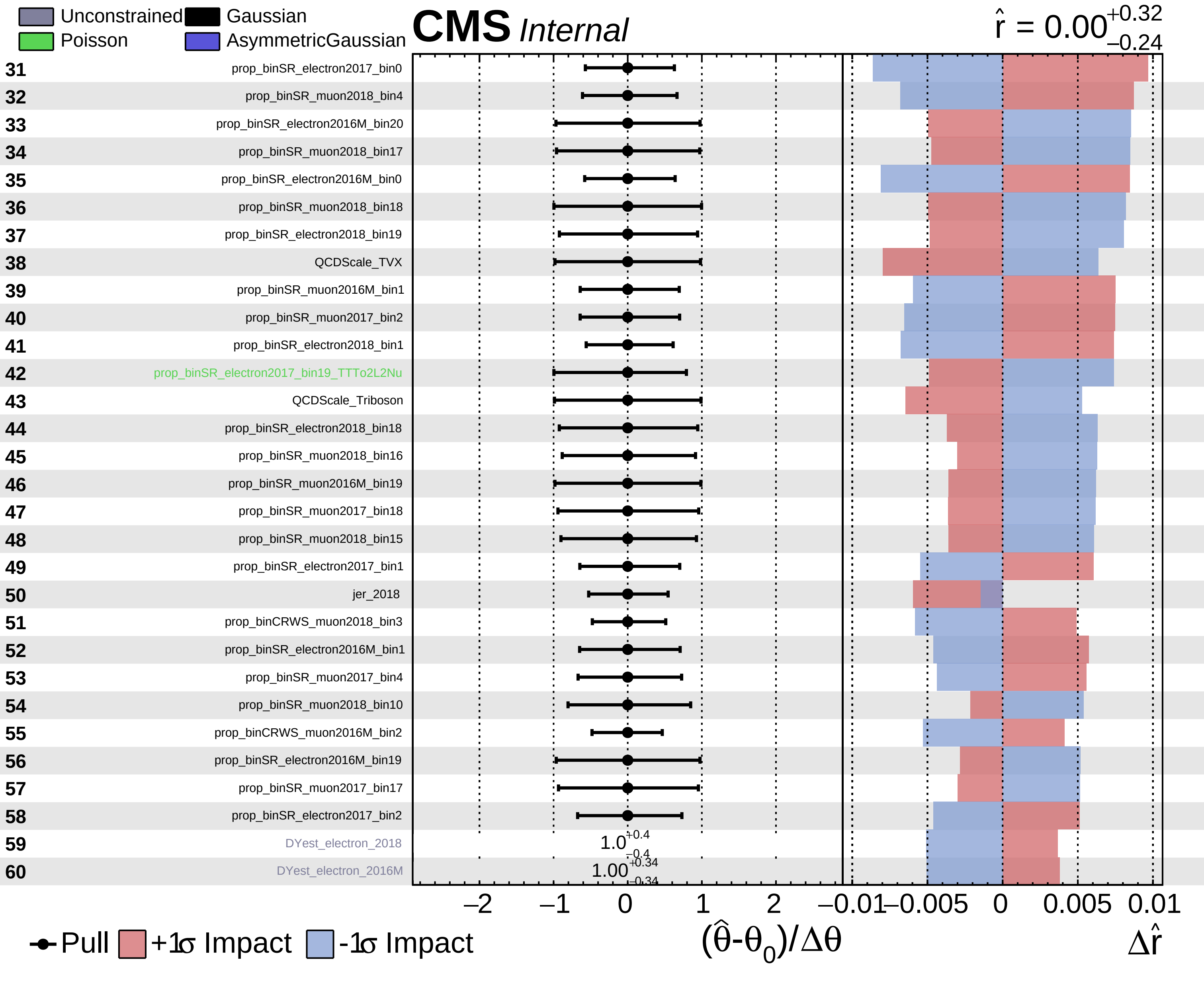
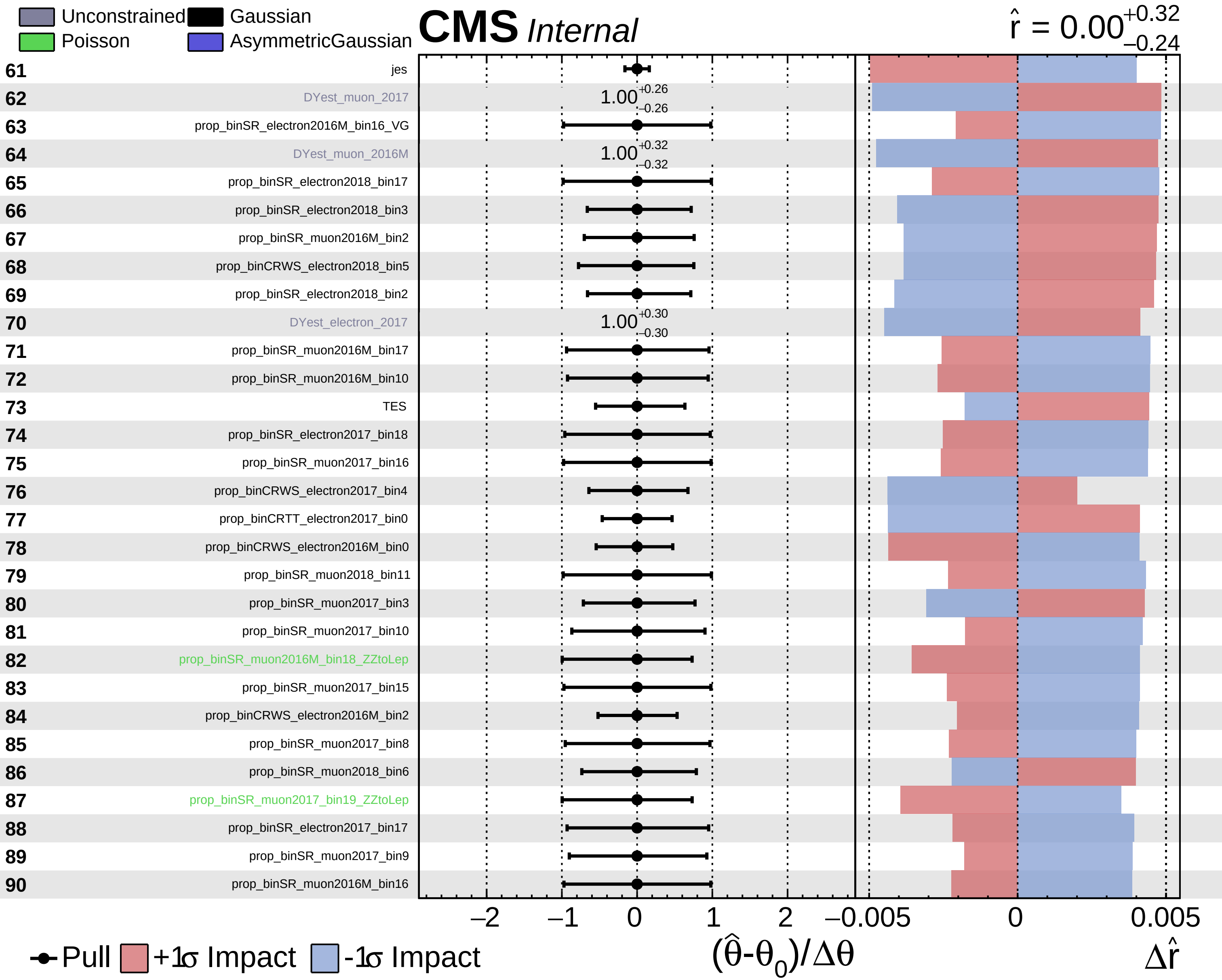
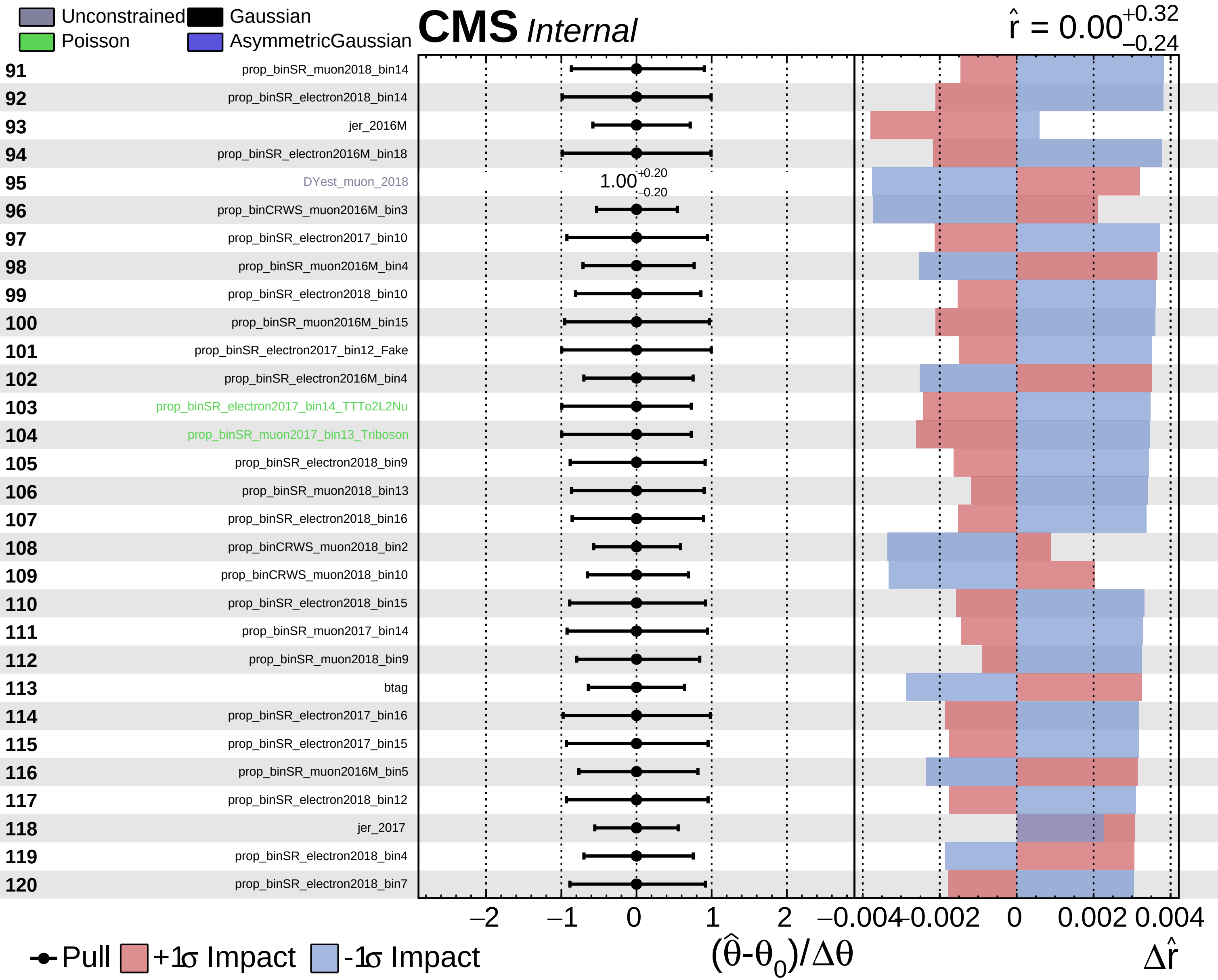


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.24
1 FRest_muon_2018 1.00 +0.10 −0.10
2 prop_binCRF_muon2018_bin0
3 FRest_muon_2017 1.00 +0.12 −0.12
4 prop_binSR_muon2018_bin0
5 FRest_muon_2016M 1.00 +0.12 −0.11
6 prop_binSR_muon2018_bin20
7 prop_binCRF_muon2017_bin0
8 FRest_electron_2018 1.00 +0.11 −0.11
9 prop_binSR_electron2017_bin14_TVX
10 prop_binCRF_muon2016M_bin0
11 prop_binSR_electron2018_bin20
12 prop_binCRF_electron2018_bin0
13 FRest_electron_2017 1.00 +0.13 −0.12
14 FRest_electron_2016M 1.00 +0.13 −0.12
15 pdf_Tot
16 prop_binSR_muon2017_bin0
17 prop_binSR_muon2017_bin20
18 prop_binCRF_electron2017_bin0
19 prop_binCRF_electron2016M_bin0
20 prop_binSR_muon2017_bin19_WrongSign
21 prop_binSR_muon2016M_bin0
22 prop_binSR_electron2018_bin0
23 prop_binSR_muon2018_bin1
24 prop_binSR_muon2016M_bin20
25 prop_binSR_muon2018_bin19
26 QCDScale_WpWpJJ_QCD
27 prop_binSR_electron2017_bin20
28 prop_binSR_muon2017_bin1
29 prop_binSR_muon2018_bin3
30 prop_binSR_muon2018_bin2
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

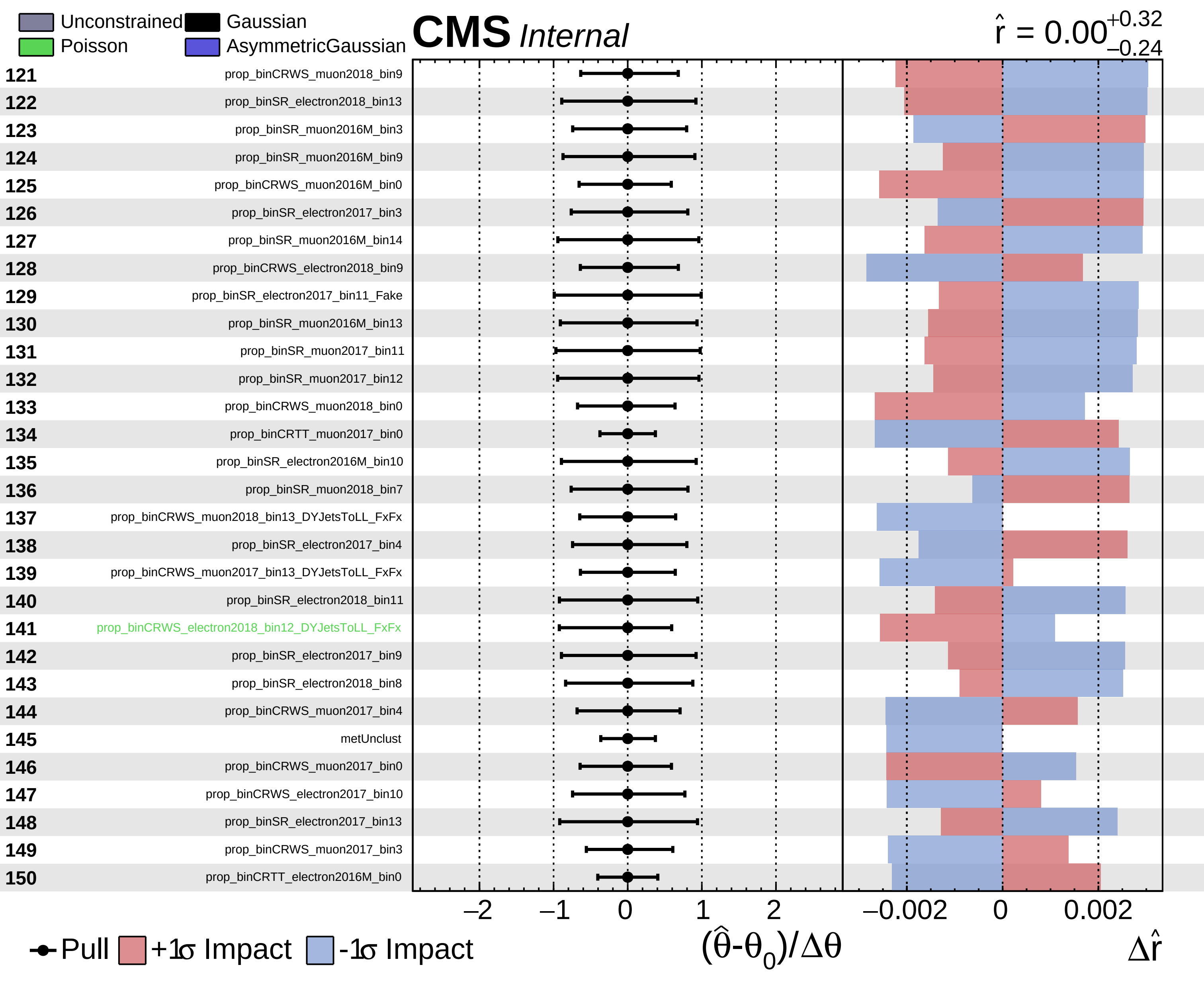


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
31 prop_binSR_electron2017_bin0
32 prop_binSR_muon2018_bin4
33 prop_binSR_electron2016M_bin20
34 prop_binSR_muon2018_bin17
35 prop_binSR_electron2016M_bin0
36 prop_binSR_muon2018_bin18
37 prop_binSR_electron2018_bin19
38 QCDScale_TVX
39 prop_binSR_muon2016M_bin1
40 prop_binSR_muon2017_bin2
41 prop_binSR_electron2018_bin1
42 prop_binSR_electron2017_bin19_TTTo2L2Nu
43 QCDScale_Triboson
44 prop_binSR_electron2018_bin18
45 prop_binSR_muon2018_bin16
46 prop_binSR_muon2016M_bin19
47 prop_binSR_muon2017_bin18
48 prop_binSR_muon2018_bin15
49 prop_binSR_electron2017_bin1
50 jer_2018
51 prop_binCRWS_muon2018_bin3
52 prop_binSR_electron2016M_bin1
53 prop_binSR_muon2017_bin4
54 prop_binSR_muon2018_bin10
55 prop_binCRWS_muon2016M_bin2
56 prop_binSR_electron2016M_bin19
57 prop_binSR_muon2017_bin17
58 prop_binSR_electron2017_bin2
59 DYest_electron_2018
60 DYest_electron_2016M
$1.0^{+0.4}_{-0.4}$
$1.00^{+0.34}_{-0.34}$
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

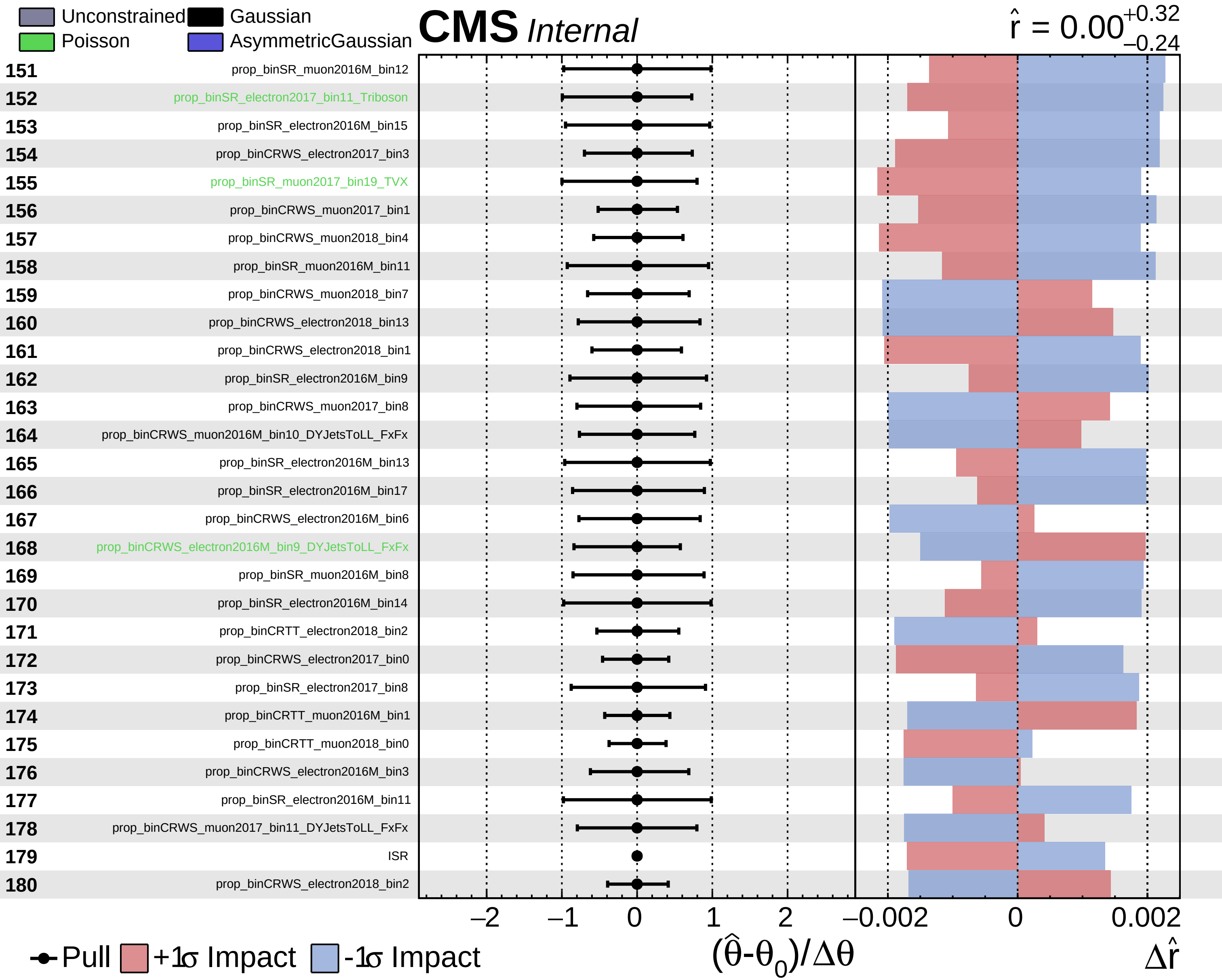

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
61 jes
62 DYest_muon_2017 $1.00^{+0.26}_{-0.26}$
63 prop_binSR_electron2016M_bin16_VG
64 DYest_muon_2016M $1.00^{+0.32}_{-0.32}$
65 prop_binSR_electron2018_bin17
66 prop_binSR_electron2018_bin3
67 prop_binSR_muon2016M_bin2
68 prop_binCRWS_electron2018_bin5
69 prop_binSR_electron2018_bin2
70 DYest_electron_2017 $1.00^{+0.30}_{-0.30}$
71 prop_binSR_muon2016M_bin17
72 prop_binSR_muon2016M_bin10
73 TES
74 prop_binSR_electron2017_bin18
75 prop_binSR_muon2017_bin16
76 prop_binCRWS_electron2017_bin4
77 prop_binCRTT_electron2017_bin0
78 prop_binCRWS_electron2016M_bin0
79 prop_binSR_muon2018_bin11
80 prop_binSR_muon2017_bin3
81 prop_binSR_muon2017_bin10
82 prop_binSR_muon2016M_bin18_ZZtoLep
83 prop_binSR_muon2017_bin15
84 prop_binCRWS_electron2016M_bin2
85 prop_binSR_muon2017_bin8
86 prop_binSR_muon2018_bin6
87 prop_binSR_muon2017_bin19_ZZtoLep
88 prop_binSR_electron2017_bin17
89 prop_binSR_muon2017_bin9
90 prop_binSR_muon2016M_bin16
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



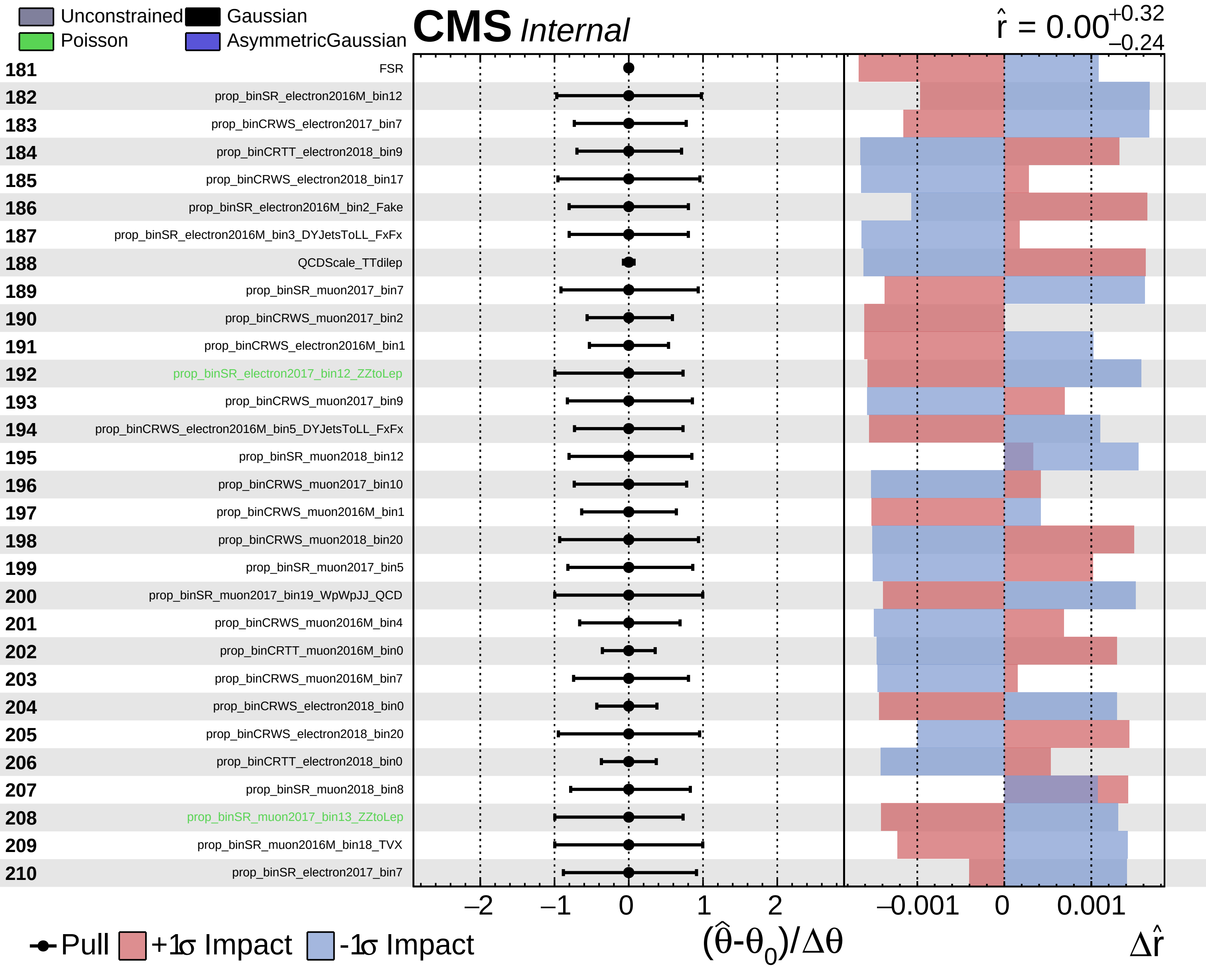
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
91 prop_binSR_muon2018_bin14
92 prop_binSR_electron2018_bin14
93 jer_2016M
94 prop_binSR_electron2016M_bin18
95 DYest_muon_2018
$1.00^{+0.20}_{-0.20}$
96 prop_binCRWS_muon2016M_bin3
97 prop_binSR_electron2017_bin10
98 prop_binSR_muon2016M_bin4
99 prop_binSR_electron2018_bin10
100 prop_binSR_muon2016M_bin15
101 prop_binSR_electron2017_bin12_Fake
102 prop_binSR_electron2016M_bin4
103 prop_binSR_electron2017_bin14_TTTo2L2Nu
104 prop_binSR_muon2017_bin13_Triboson
105 prop_binSR_electron2018_bin9
106 prop_binSR_muon2018_bin13
107 prop_binSR_electron2018_bin16
108 prop_binCRWS_muon2018_bin2
109 prop_binCRWS_muon2018_bin10
110 prop_binSR_electron2018_bin15
111 prop_binSR_muon2017_bin14
112 prop_binSR_muon2018_bin9
113 btag
114 prop_binSR_electron2017_bin16
115 prop_binSR_electron2017_bin15
116 prop_binSR_muon2016M_bin5
117 prop_binSR_electron2018_bin12
118 jer_2017
119 prop_binSR_electron2018_bin4
120 prop_binSR_electron2018_bin7
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



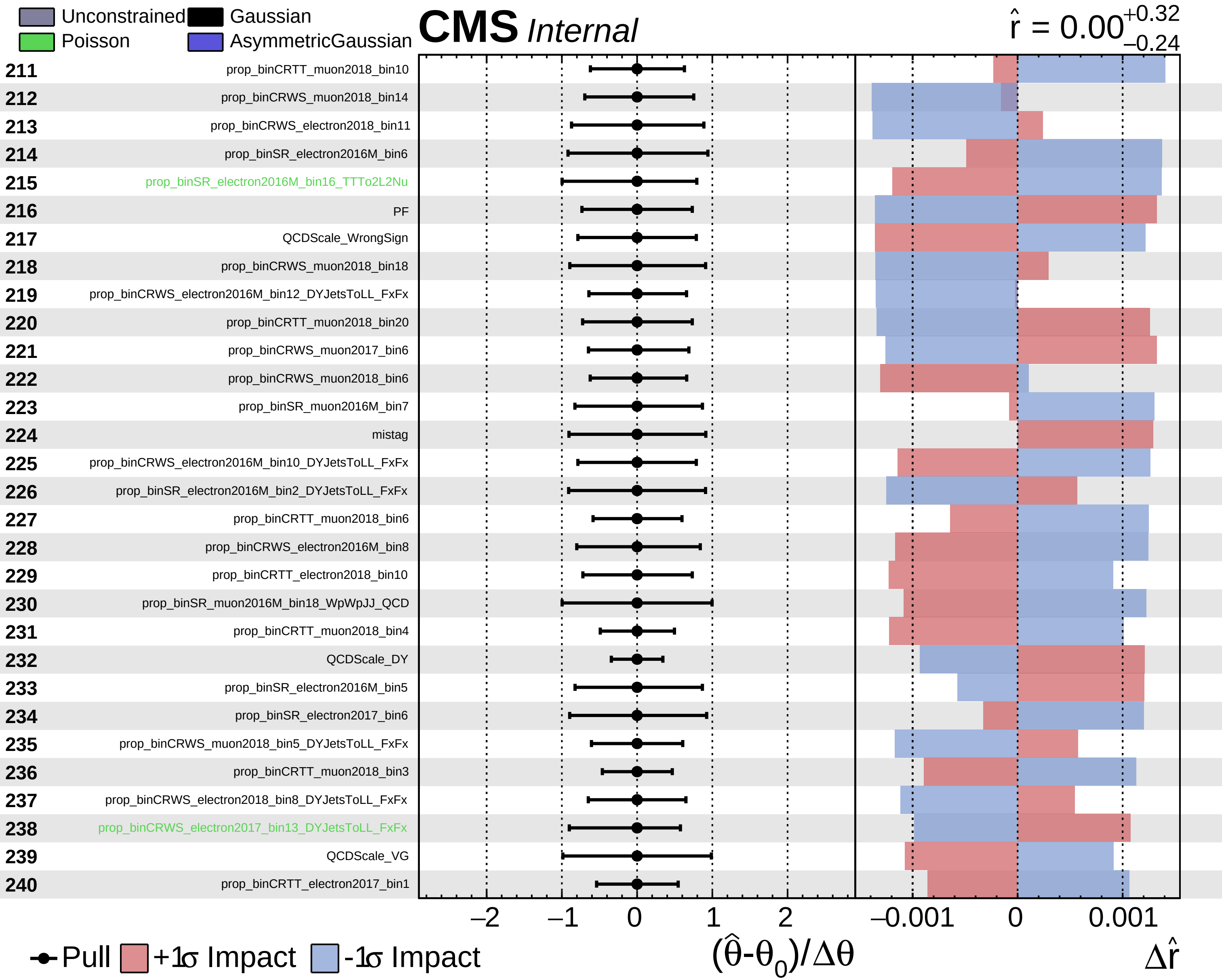
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
121 prop_binCRWS_muon2018_bin9
122 prop_binSR_electron2018_bin13
123 prop_binSR_muon2016M_bin3
124 prop_binSR_muon2016M_bin9
125 prop_binCRWS_muon2016M_bin0
126 prop_binSR_electron2017_bin3
127 prop_binSR_muon2016M_bin14
128 prop_binCRWS_electron2018_bin9
129 prop_binSR_electron2017_bin11_Fake
130 prop_binSR_muon2016M_bin13
131 prop_binSR_muon2017_bin11
132 prop_binSR_muon2017_bin12
133 prop_binCRWS_muon2018_bin0
134 prop_binCRTT_muon2017_bin0
135 prop_binSR_electron2016M_bin10
136 prop_binSR_muon2018_bin7
137 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
138 prop_binSR_electron2017_bin4
139 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
140 prop_binSR_electron2018_bin11
141 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
142 prop_binSR_electron2017_bin9
143 prop_binSR_electron2018_bin8
144 prop_binCRWS_muon2017_bin4
145 metUnclust
146 prop_binCRWS_muon2017_bin0
147 prop_binCRWS_electron2017_bin10
148 prop_binSR_electron2017_bin13
149 prop_binCRWS_muon2017_bin3
150 prop_binCRTT_electron2016M_bin0
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



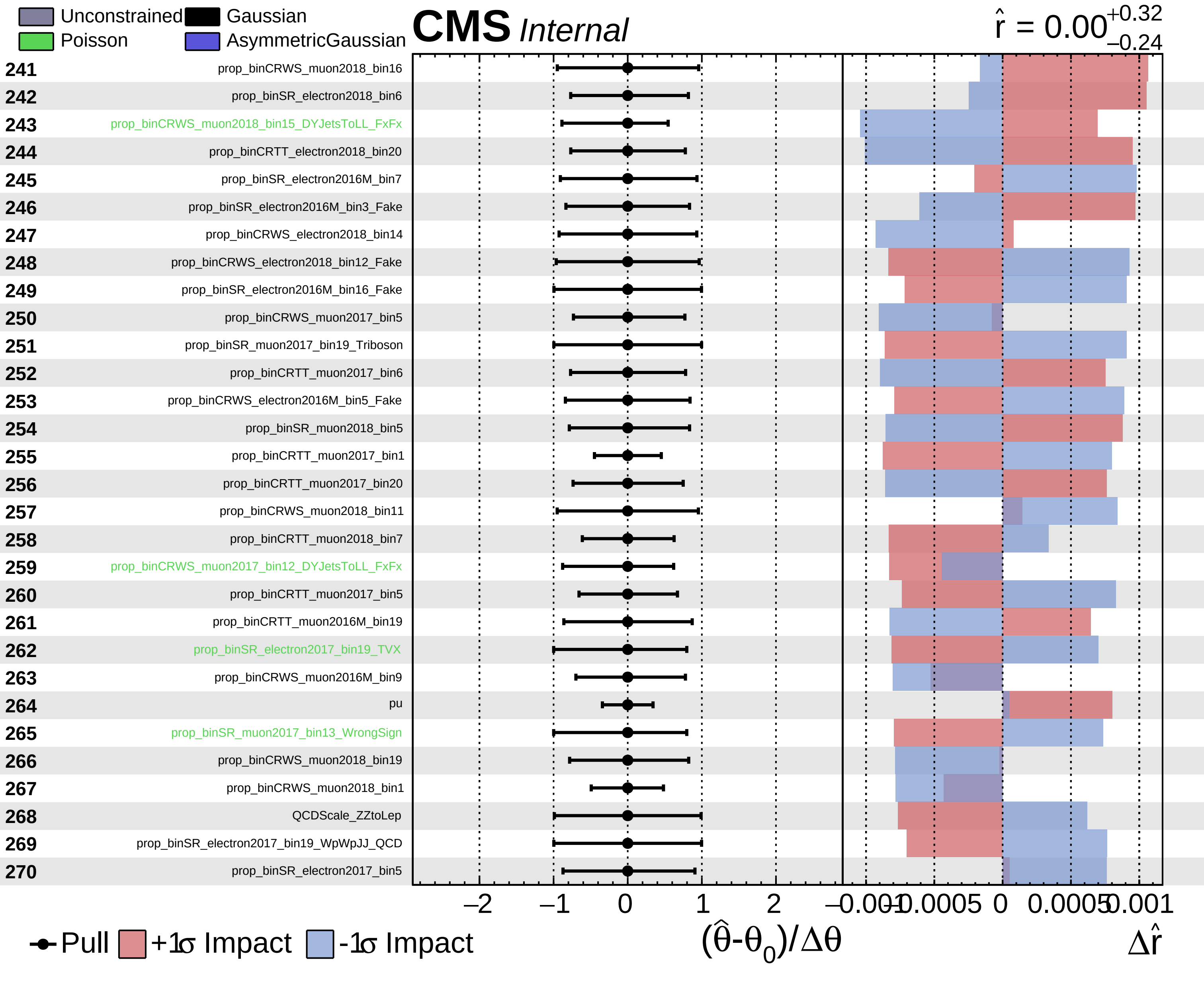
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
151 prop_binSR_muon2016M_bin12
152 prop_binSR_electron2017_bin11_Triboson
153 prop_binSR_electron2016M_bin15
154 prop_binCRWS_electron2017_bin3
155 prop_binSR_muon2017_bin19_TVX
156 prop_binCRWS_muon2017_bin1
157 prop_binCRWS_muon2018_bin4
158 prop_binSR_muon2016M_bin11
159 prop_binCRWS_muon2018_bin7
160 prop_binCRWS_electron2018_bin13
161 prop_binCRWS_electron2018_bin1
162 prop_binSR_electron2016M_bin9
163 prop_binCRWS_muon2017_bin8
164 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
165 prop_binSR_electron2016M_bin13
166 prop_binSR_electron2016M_bin17
167 prop_binCRWS_electron2016M_bin6
168 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
169 prop_binSR_muon2016M_bin8
170 prop_binSR_electron2016M_bin14
171 prop_binCRTT_electron2018_bin2
172 prop_binCRWS_electron2017_bin0
173 prop_binSR_electron2017_bin8
174 prop_binCRTT_muon2016M_bin1
175 prop_binCRTT_muon2018_bin0
176 prop_binCRWS_electron2016M_bin3
177 prop_binSR_electron2016M_bin11
178 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
179 ISR
180 prop_binCRWS_electron2018_bin2
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



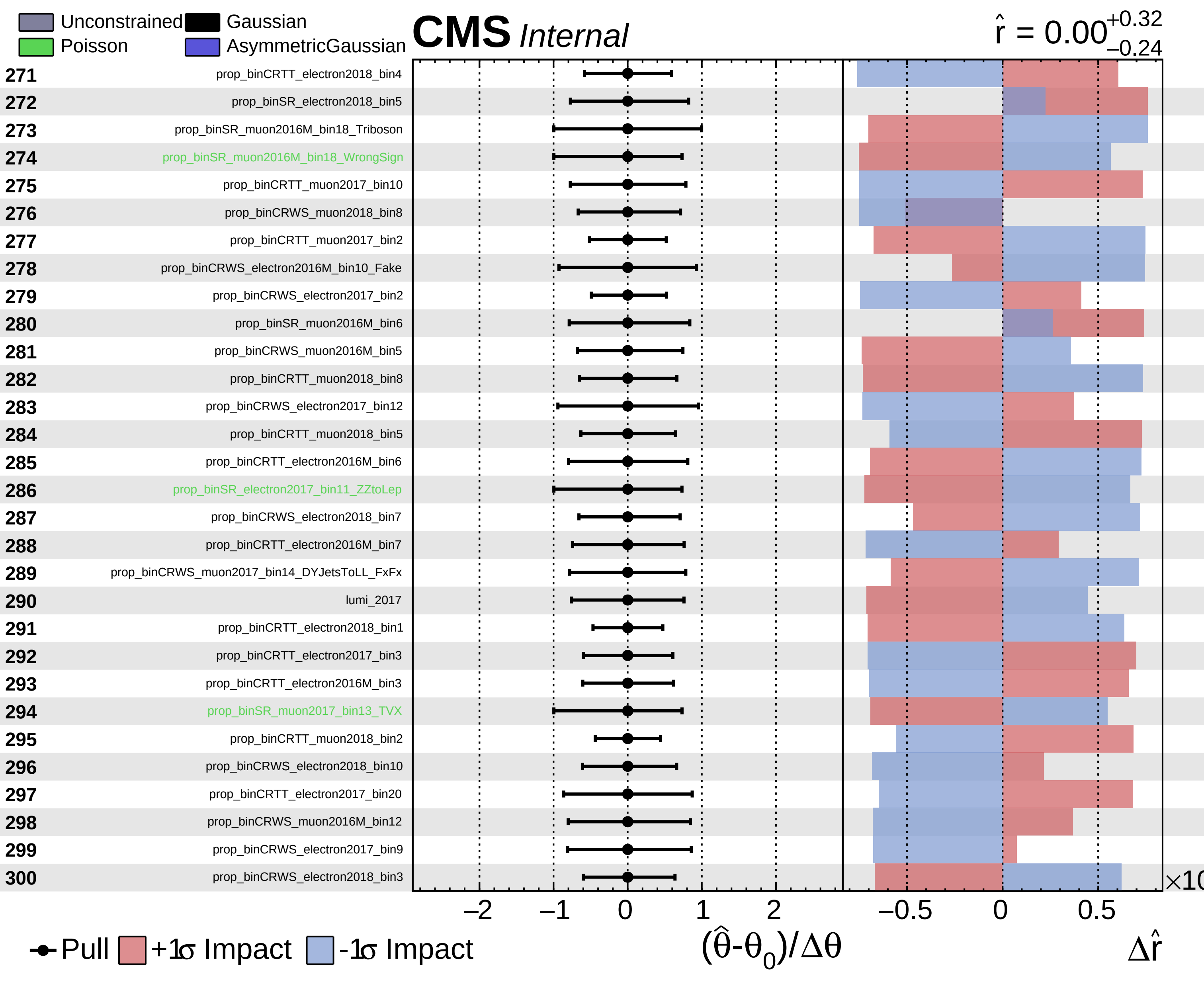
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
181 FSR
182 prop_binSR_electron2016M_bin12
183 prop_binCRWS_electron2017_bin7
184 prop_binCRTT_electron2018_bin9
185 prop_binCRWS_electron2018_bin17
186 prop_binSR_electron2016M_bin2_Fake
187 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
188 QCDScale_TTdilep
189 prop_binSR_muon2017_bin7
190 prop_binCRWS_muon2017_bin2
191 prop_binCRWS_electron2016M_bin1
192 prop_binSR_electron2017_bin12_ZZtoLep
193 prop_binCRWS_muon2017_bin9
194 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
195 prop_binSR_muon2018_bin12
196 prop_binCRWS_muon2017_bin10
197 prop_binCRWS_muon2016M_bin1
198 prop_binCRWS_muon2018_bin20
199 prop_binSR_muon2017_bin5
200 prop_binSR_muon2017_bin19_WpWpJJ_QCD
201 prop_binCRWS_muon2016M_bin4
202 prop_binCRTT_muon2016M_bin0
203 prop_binCRWS_muon2016M_bin7
204 prop_binCRWS_electron2018_bin0
205 prop_binCRWS_electron2018_bin20
206 prop_binCRTT_electron2018_bin0
207 prop_binSR_muon2018_bin8
208 prop_binSR_muon2017_bin13_ZZtoLep
209 prop_binSR_muon2016M_bin18_TVX
210 prop_binSR_electron2017_bin7
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



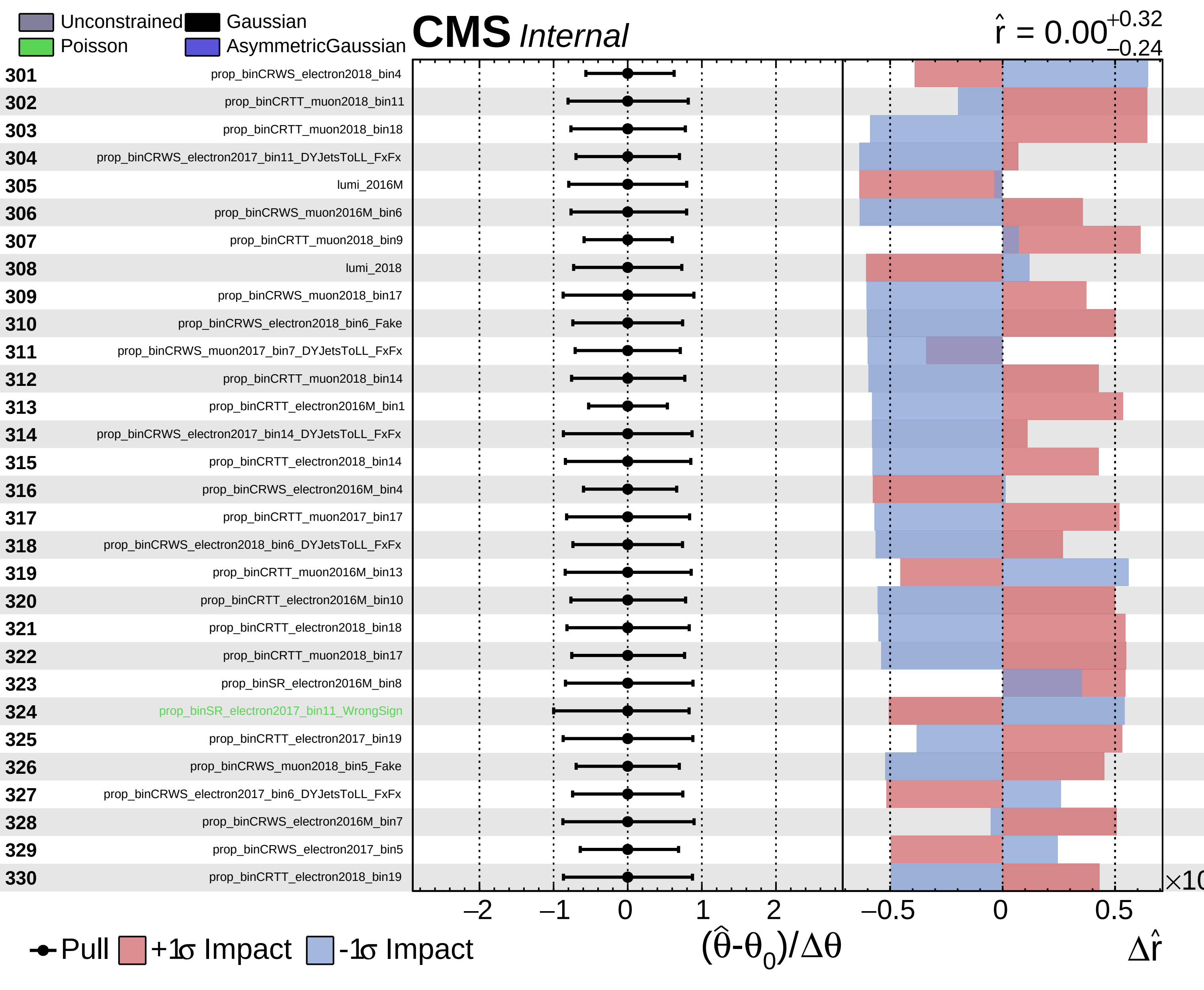
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
211 prop_binCRTT_muon2018_bin10
212 prop_binCRWS_muon2018_bin14
213 prop_binCRWS_electron2018_bin11
214 prop_binSR_electron2016M_bin6
215 prop_binSR_electron2016M_bin16_TTTo2L2Nu
216 PF
217 QCDScale_WrongSign
218 prop_binCRWS_muon2018_bin18
219 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
220 prop_binCRTT_muon2018_bin20
221 prop_binCRWS_muon2017_bin6
222 prop_binCRWS_muon2018_bin6
223 prop_binSR_muon2016M_bin7
224 mistag
225 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
226 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
227 prop_binCRTT_muon2018_bin6
228 prop_binCRWS_electron2016M_bin8
229 prop_binCRTT_electron2018_bin10
230 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
231 prop_binCRTT_muon2018_bin4
232 QCDScale_DY
233 prop_binSR_electron2016M_bin5
234 prop_binSR_electron2017_bin6
235 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
236 prop_binCRTT_muon2018_bin3
237 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
238 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
239 QCDScale_VG
240 prop_binCRTT_electron2017_bin1
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



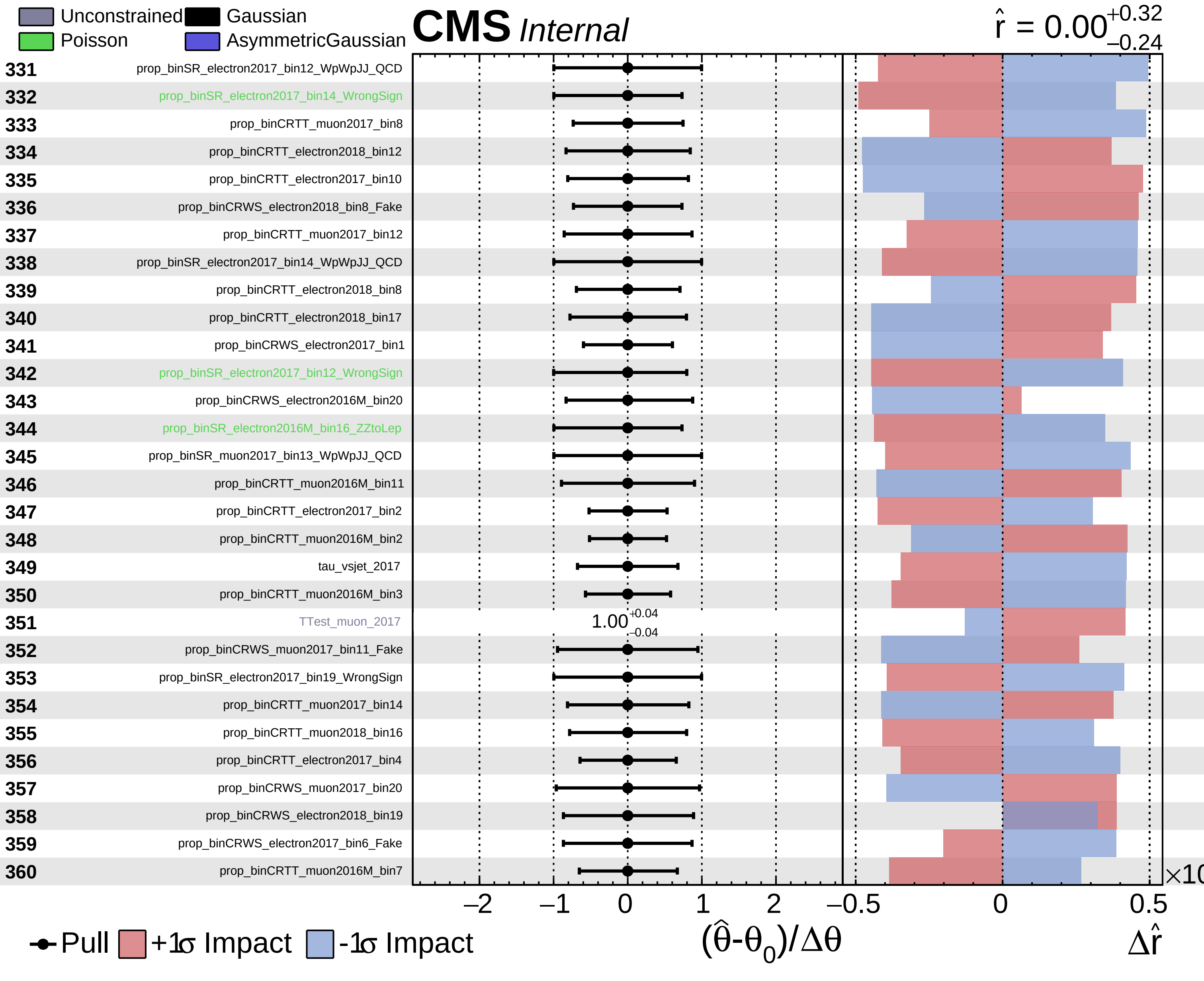
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.24
241 prop_binCRWS_muon2018_bin16
242 prop_binSR_electron2018_bin6
243 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
244 prop_binCRTT_electron2018_bin20
245 prop_binSR_electron2016M_bin7
246 prop_binSR_electron2016M_bin3_Fake
247 prop_binCRWS_electron2018_bin14
248 prop_binCRWS_electron2018_bin12_Fake
249 prop_binSR_electron2016M_bin16_Fake
250 prop_binCRWS_muon2017_bin5
251 prop_binSR_muon2017_bin19_Triboson
252 prop_binCRTT_muon2017_bin6
253 prop_binCRWS_electron2016M_bin5_Fake
254 prop_binSR_muon2018_bin5
255 prop_binCRTT_muon2017_bin1
256 prop_binCRTT_muon2017_bin20
257 prop_binCRWS_muon2018_bin11
258 prop_binCRTT_muon2018_bin7
259 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
260 prop_binCRTT_muon2017_bin5
261 prop_binCRTT_muon2016M_bin19
262 prop_binSR_electron2017_bin19_TVX
263 prop_binCRWS_muon2016M_bin9
264 pu
265 prop_binSR_muon2017_bin13_WrongSign
266 prop_binCRWS_muon2018_bin19
267 prop_binCRWS_muon2018_bin1
268 QCDScale_ZZtoLep
269 prop_binSR_electron2017_bin19_WpWpJJ_QCD
270 prop_binSR_electron2017_bin5
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

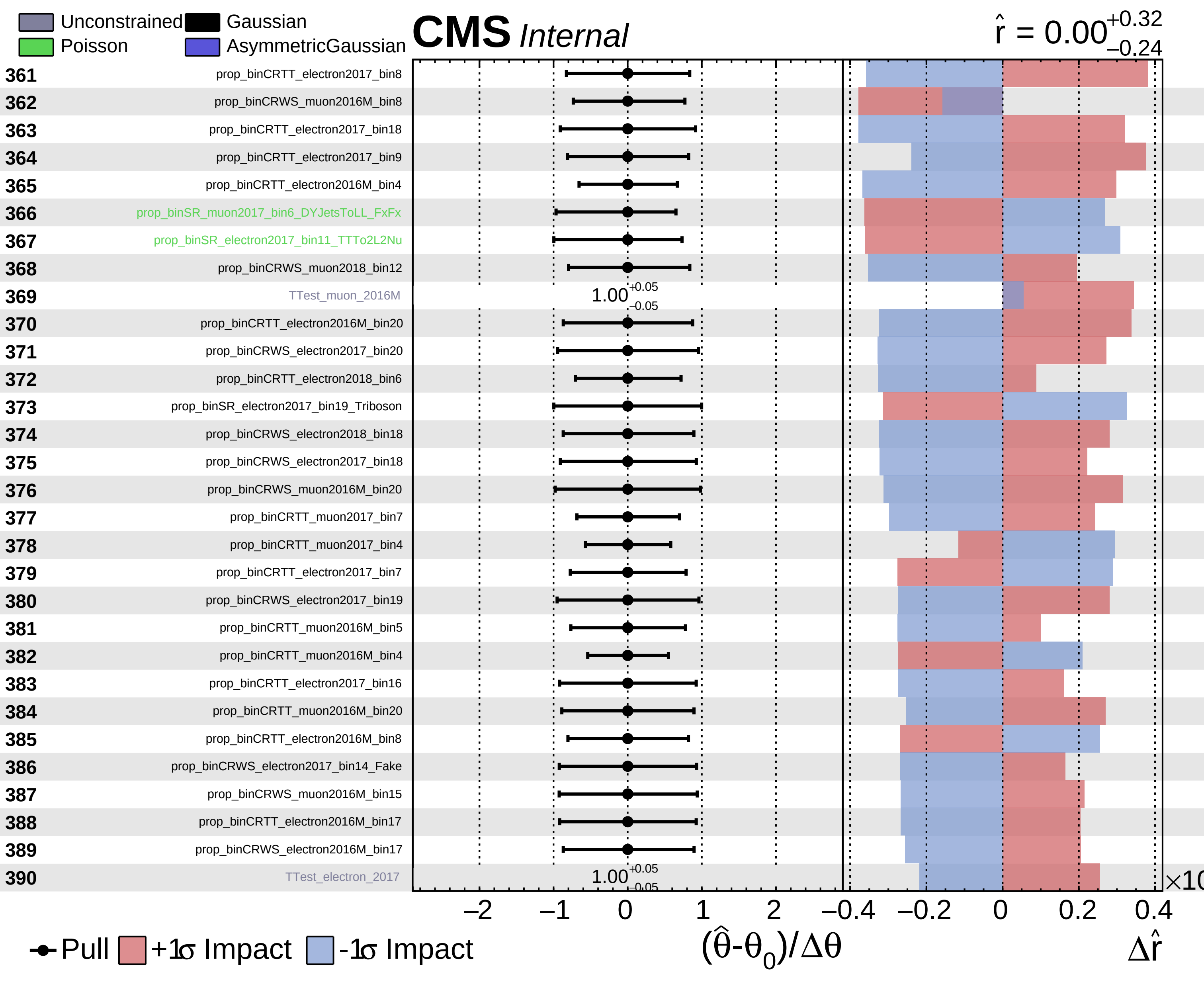


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
271 prop_binCRTT_electron2018_bin4
272 prop_binSR_electron2018_bin5
273 prop_binSR_muon2016M_bin18_Triboson
274 prop_binSR_muon2016M_bin18_WrongSign
275 prop_binCRTT_muon2017_bin10
276 prop_binCRWS_muon2018_bin8
277 prop_binCRTT_muon2017_bin2
278 prop_binCRWS_electron2016M_bin10_Fake
279 prop_binCRWS_electron2017_bin2
280 prop_binSR_muon2016M_bin6
281 prop_binCRWS_muon2016M_bin5
282 prop_binCRTT_muon2018_bin8
283 prop_binCRWS_electron2017_bin12
284 prop_binCRTT_muon2018_bin5
285 prop_binCRTT_electron2016M_bin6
286 prop_binSR_electron2017_bin11_ZZtoLep
287 prop_binCRWS_electron2018_bin7
288 prop_binCRTT_electron2016M_bin7
289 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
290 lumi_2017
291 prop_binCRTT_electron2018_bin1
292 prop_binCRTT_electron2017_bin3
293 prop_binCRTT_electron2016M_bin3
294 prop_binSR_muon2017_bin13_TVX
295 prop_binCRTT_muon2018_bin2
296 prop_binCRWS_electron2018_bin10
297 prop_binCRTT_electron2017_bin20
298 prop_binCRWS_muon2016M_bin12
299 prop_binCRWS_electron2017_bin9
300 prop_binCRWS_electron2018_bin3
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

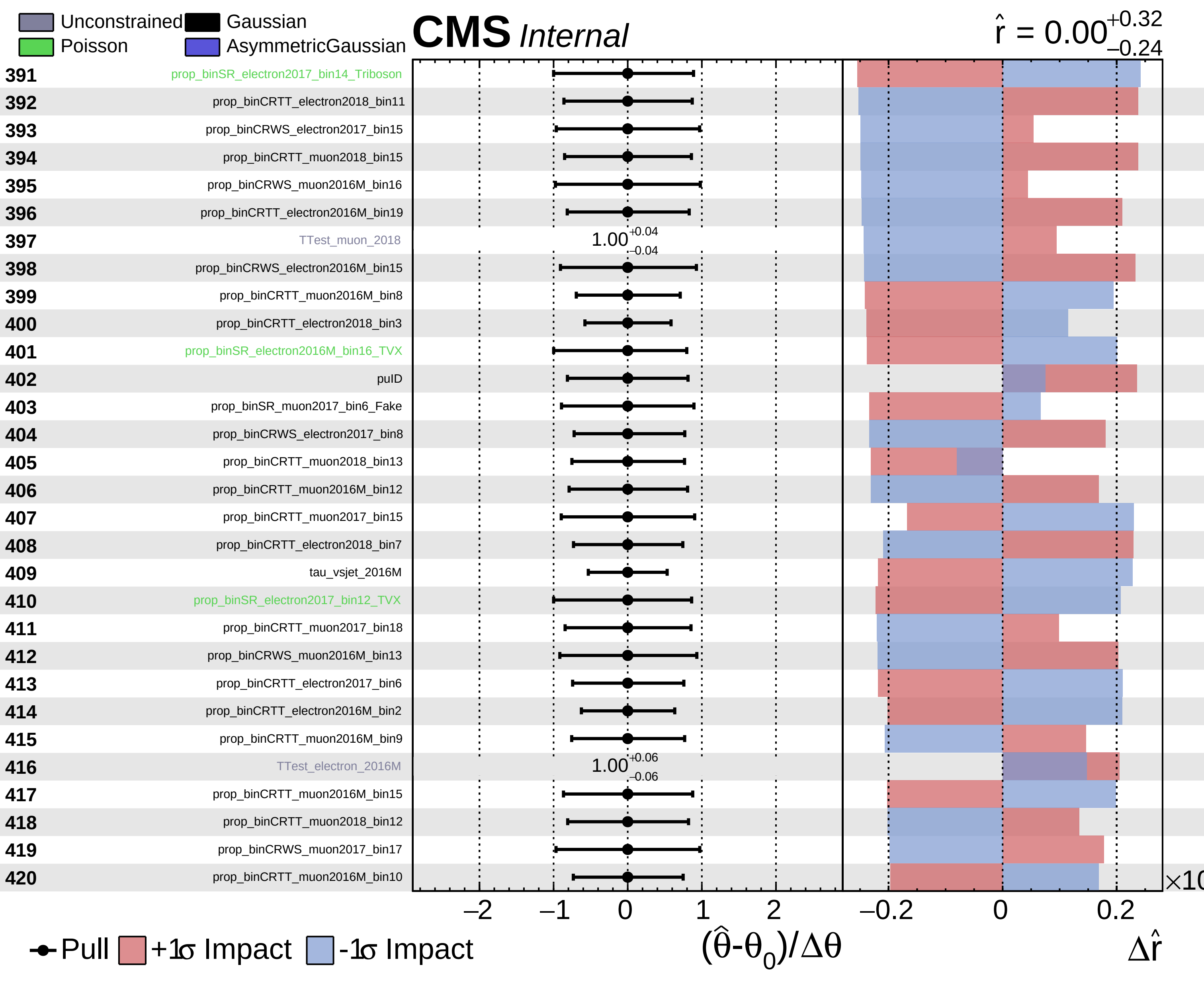


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.24
301 prop_binCRWS_electron2018_bin4
302 prop_binCRTT_muon2018_bin11
303 prop_binCRTT_muon2018_bin18
304 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
305 lumi_2016M
306 prop_binCRWS_muon2016M_bin6
307 prop_binCRTT_muon2018_bin9
308 lumi_2018
309 prop_binCRWS_muon2018_bin17
310 prop_binCRWS_electron2018_bin6_Fake
311 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
312 prop_binCRTT_muon2018_bin14
313 prop_binCRTT_electron2016M_bin1
314 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
315 prop_binCRTT_electron2018_bin14
316 prop_binCRWS_electron2016M_bin4
317 prop_binCRTT_muon2017_bin17
318 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
319 prop_binCRTT_muon2016M_bin13
320 prop_binCRTT_electron2016M_bin10
321 prop_binCRTT_electron2018_bin18
322 prop_binCRTT_muon2018_bin17
323 prop_binSR_electron2016M_bin8
324 prop_binSR_electron2017_bin11_WrongSign
325 prop_binCRTT_electron2017_bin19
326 prop_binCRWS_muon2018_bin5_Fake
327 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
328 prop_binCRWS_electron2016M_bin7
329 prop_binCRWS_electron2017_bin5
330 prop_binCRTT_electron2018_bin19
×10
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

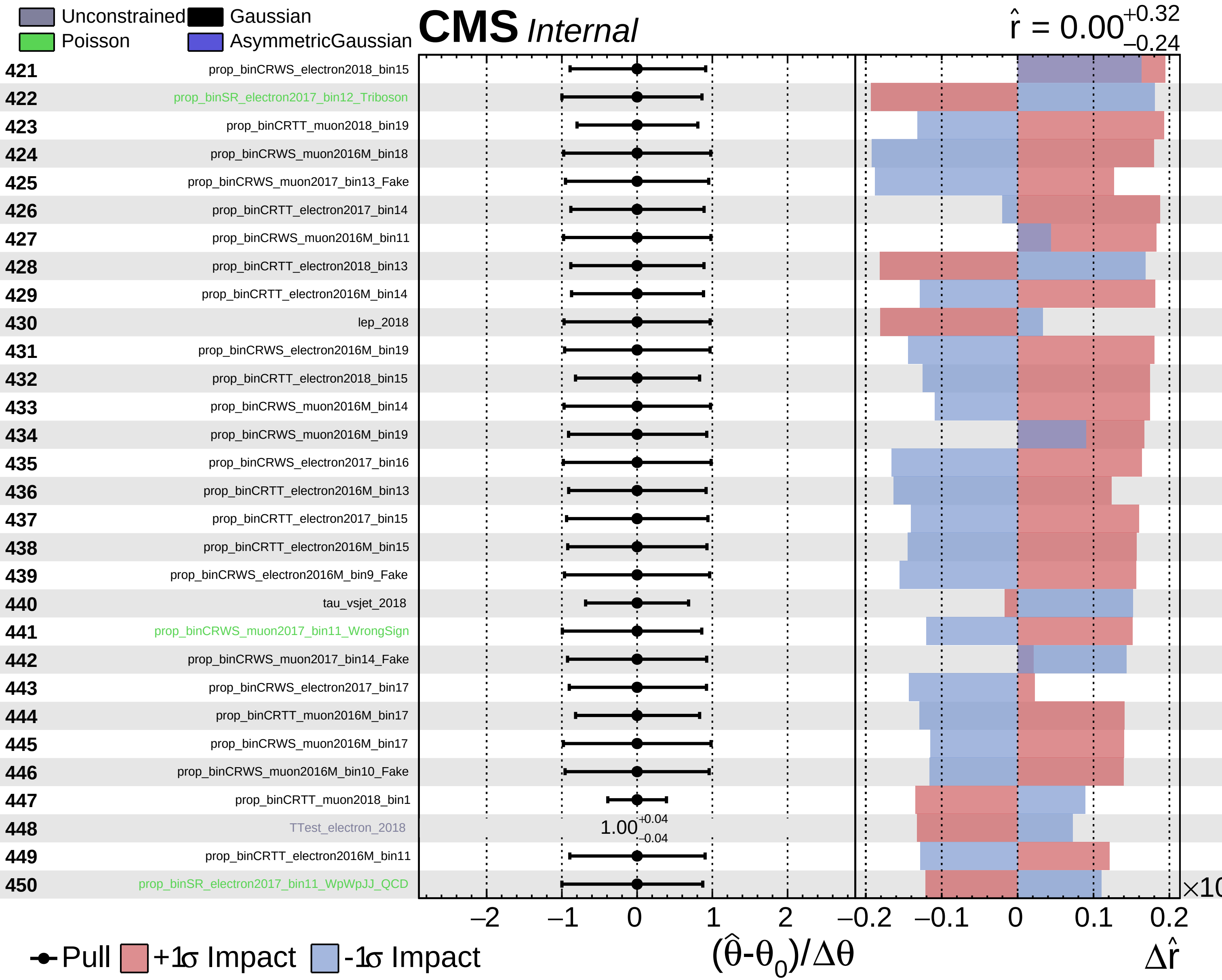

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
331 prop_binSR_electron2017_bin12_WpWpJJ_QCD
332 prop_binSR_electron2017_bin14_WrongSign
333 prop_binCRTT_muon2017_bin8
334 prop_binCRTT_electron2018_bin12
335 prop_binCRTT_electron2017_bin10
336 prop_binCRWS_electron2018_bin8_Fake
337 prop_binCRTT_muon2017_bin12
338 prop_binSR_electron2017_bin14_WpWpJJ_QCD
339 prop_binCRTT_electron2018_bin8
340 prop_binCRTT_electron2018_bin17
341 prop_binCRWS_electron2017_bin1
342 prop_binSR_electron2017_bin12_WrongSign
343 prop_binCRWS_electron2016M_bin20
344 prop_binSR_electron2016M_bin16_ZZtoLep
345 prop_binSR_muon2017_bin13_WpWpJJ_QCD
346 prop_binCRTT_muon2016M_bin11
347 prop_binCRTT_electron2017_bin2
348 prop_binCRTT_muon2016M_bin2
349 tau_vsjet_2017
350 prop_binCRTT_muon2016M_bin3
351 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
352 prop_binCRWS_muon2017_bin11_Fake
353 prop_binSR_electron2017_bin19_WrongSign
354 prop_binCRTT_muon2017_bin14
355 prop_binCRTT_muon2018_bin16
356 prop_binCRTT_electron2017_bin4
357 prop_binCRWS_muon2017_bin20
358 prop_binCRWS_electron2018_bin19
359 prop_binCRWS_electron2017_bin6_Fake
360 prop_binCRTT_muon2016M_bin7
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



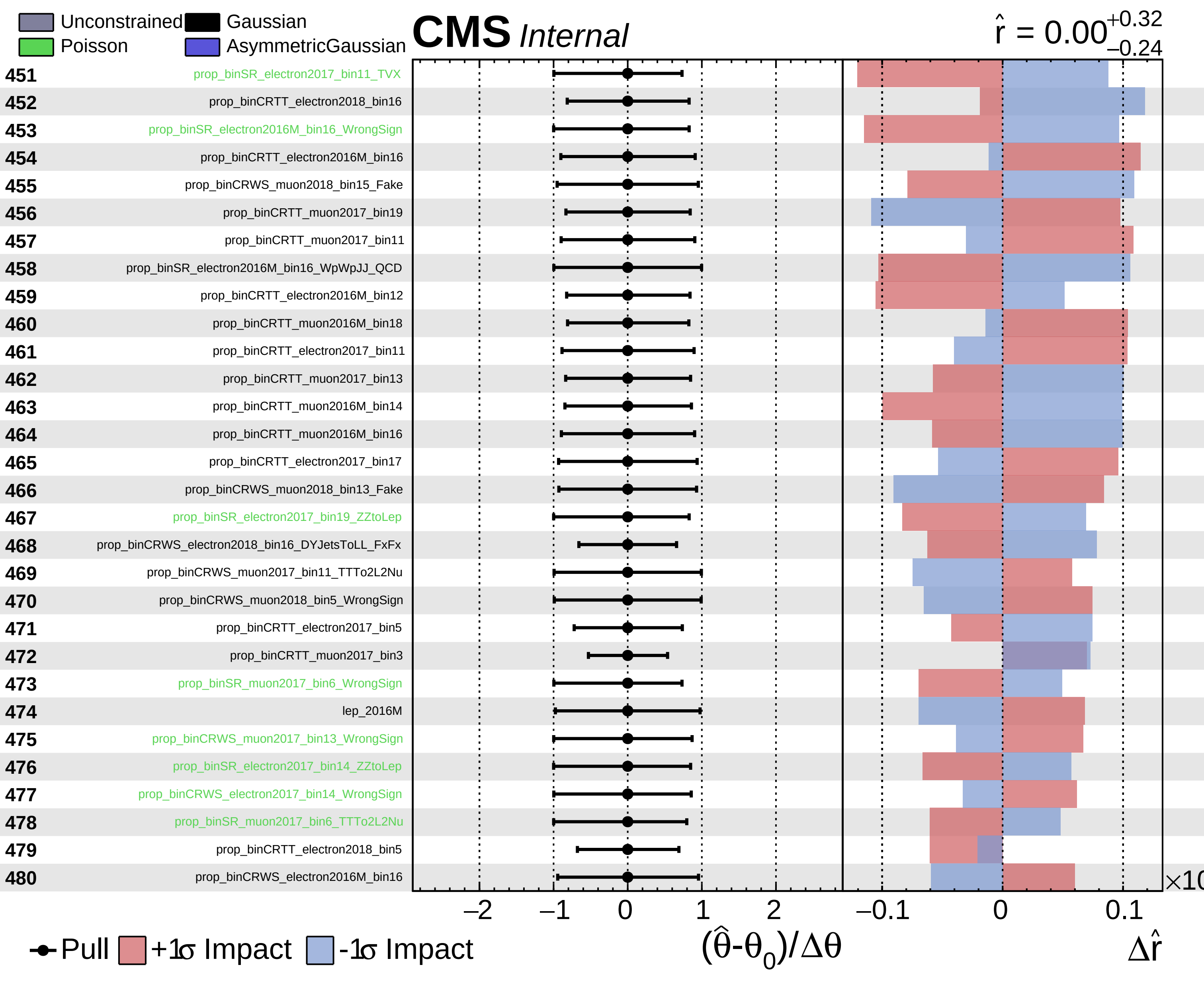
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
361 prop_binCRTT_electron2017_bin8
362 prop_binCRWS_muon2016M_bin8
363 prop_binCRTT_electron2017_bin18
364 prop_binCRTT_electron2017_bin9
365 prop_binCRTT_electron2016M_bin4
366 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
367 prop_binSR_electron2017_bin11_TTTo2L2Nu
368 prop_binCRWS_muon2018_bin12
369 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
370 prop_binCRTT_electron2016M_bin20
371 prop_binCRWS_electron2017_bin20
372 prop_binCRTT_electron2018_bin6
373 prop_binSR_electron2017_bin19_Triboson
374 prop_binCRWS_electron2018_bin18
375 prop_binCRWS_electron2017_bin18
376 prop_binCRWS_muon2016M_bin20
377 prop_binCRTT_muon2017_bin7
378 prop_binCRTT_muon2017_bin4
379 prop_binCRTT_electron2017_bin7
380 prop_binCRWS_electron2017_bin19
381 prop_binCRTT_muon2016M_bin5
382 prop_binCRTT_muon2016M_bin4
383 prop_binCRTT_electron2017_bin16
384 prop_binCRTT_muon2016M_bin20
385 prop_binCRTT_electron2016M_bin8
386 prop_binCRWS_electron2017_bin14_Fake
387 prop_binCRWS_muon2016M_bin15
388 prop_binCRTT_electron2016M_bin17
389 prop_binCRWS_electron2016M_bin17
390 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



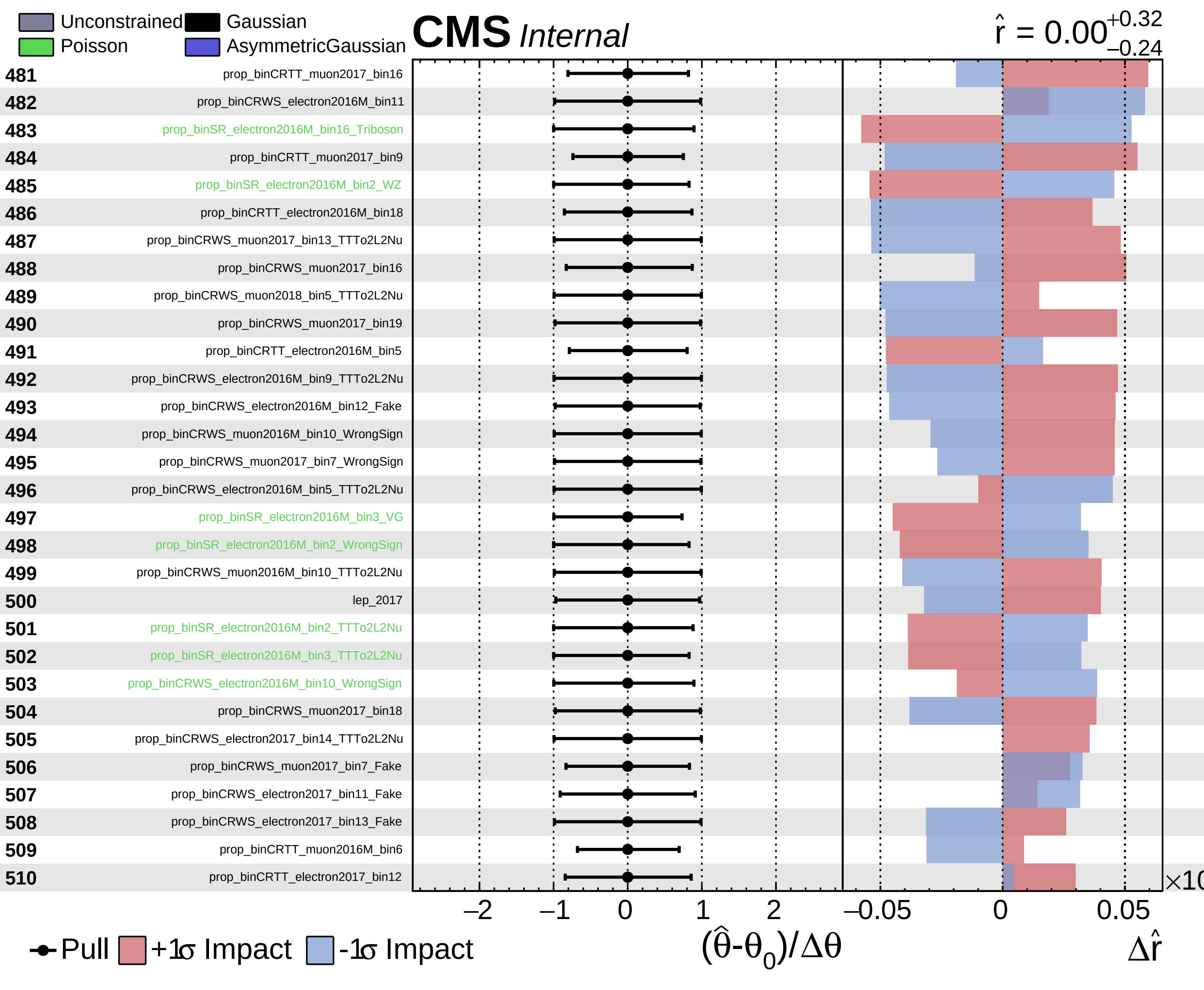
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
391 prop_binSR_electron2017_bin14_Triboson
392 prop_binCRTT_electron2018_bin11
393 prop_binCRWS_electron2017_bin15
394 prop_binCRTT_muon2018_bin15
395 prop_binCRWS_muon2016M_bin16
396 prop_binCRTT_electron2016M_bin19
397 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
398 prop_binCRWS_electron2016M_bin15
399 prop_binCRTT_muon2016M_bin8
400 prop_binCRTT_electron2018_bin3
401 prop_binSR_electron2016M_bin16_TVX
402 puID
403 prop_binSR_muon2017_bin6_Fake
404 prop_binCRWS_electron2017_bin8
405 prop_binCRTT_muon2018_bin13
406 prop_binCRTT_muon2016M_bin12
407 prop_binCRTT_muon2017_bin15
408 prop_binCRTT_electron2018_bin7
409 tau_vsjet_2016M
410 prop_binSR_electron2017_bin12_TVX
411 prop_binCRTT_muon2017_bin18
412 prop_binCRWS_muon2016M_bin13
413 prop_binCRTT_electron2017_bin6
414 prop_binCRTT_electron2016M_bin2
415 prop_binCRTT_muon2016M_bin9
416 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
417 prop_binCRTT_muon2016M_bin15
418 prop_binCRTT_muon2018_bin12
419 prop_binCRWS_muon2017_bin17
420 prop_binCRTT_muon2016M_bin10
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



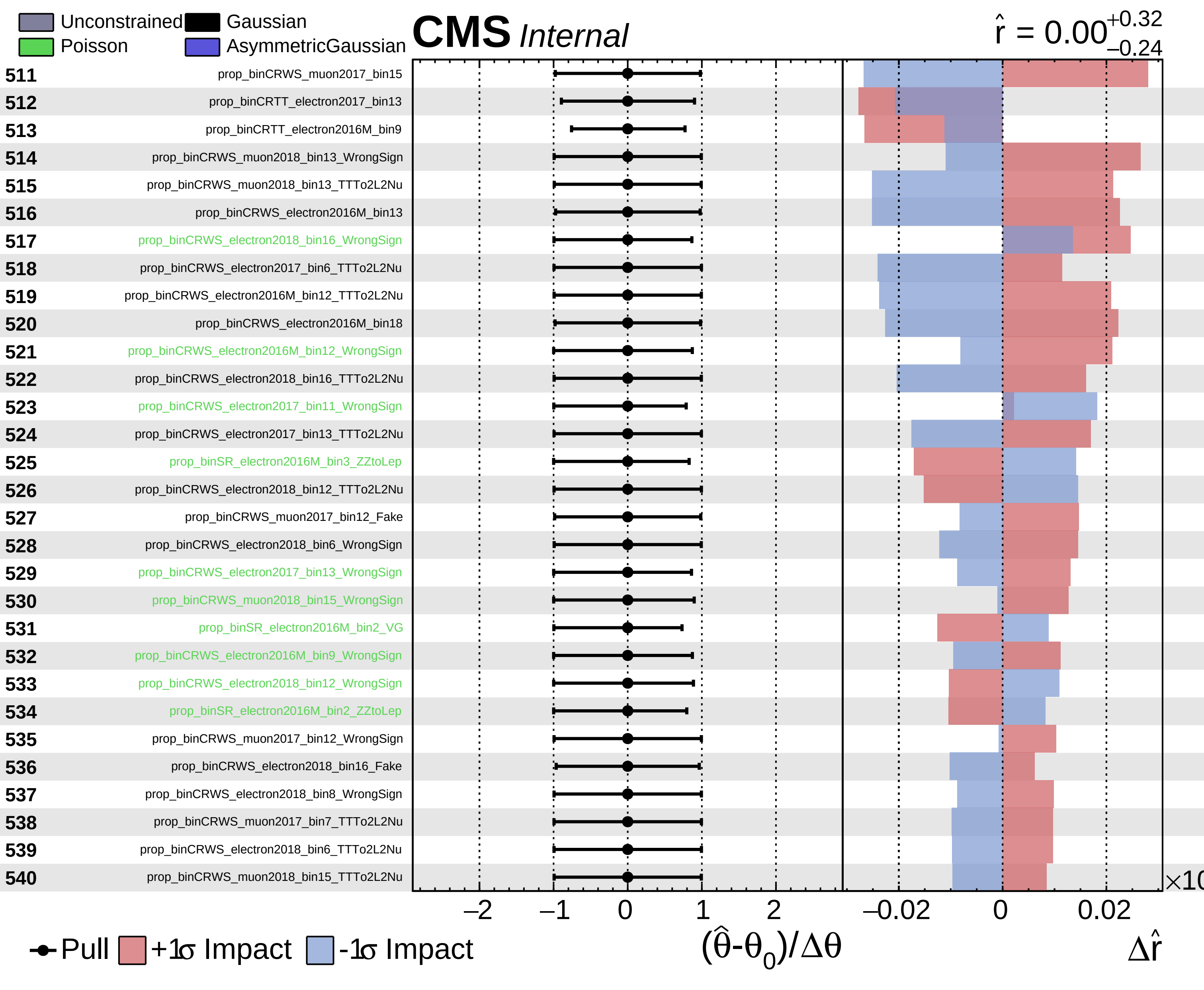
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
421 prop_binCRWS_electron2018_bin15
422 prop_binSR_electron2017_bin12_Triboson
423 prop_binCRTT_muon2018_bin19
424 prop_binCRWS_muon2016M_bin18
425 prop_binCRWS_muon2017_bin13_Fake
426 prop_binCRTT_electron2017_bin14
427 prop_binCRWS_muon2016M_bin11
428 prop_binCRTT_electron2018_bin13
429 prop_binCRTT_electron2016M_bin14
430 lep_2018
431 prop_binCRWS_electron2016M_bin19
432 prop_binCRTT_electron2018_bin15
433 prop_binCRWS_muon2016M_bin14
434 prop_binCRWS_muon2016M_bin19
435 prop_binCRWS_electron2017_bin16
436 prop_binCRTT_electron2016M_bin13
437 prop_binCRTT_electron2017_bin15
438 prop_binCRTT_electron2016M_bin15
439 prop_binCRWS_electron2016M_bin9_Fake
440 tau_vsjet_2018
441 prop_binCRWS_muon2017_bin11_WrongSign
442 prop_binCRWS_muon2017_bin14_Fake
443 prop_binCRWS_electron2017_bin17
444 prop_binCRTT_muon2016M_bin17
445 prop_binCRWS_muon2016M_bin17
446 prop_binCRWS_muon2016M_bin10_Fake
447 prop_binCRTT_muon2018_bin1
448 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
449 prop_binCRTT_electron2016M_bin11
450 prop_binSR_electron2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



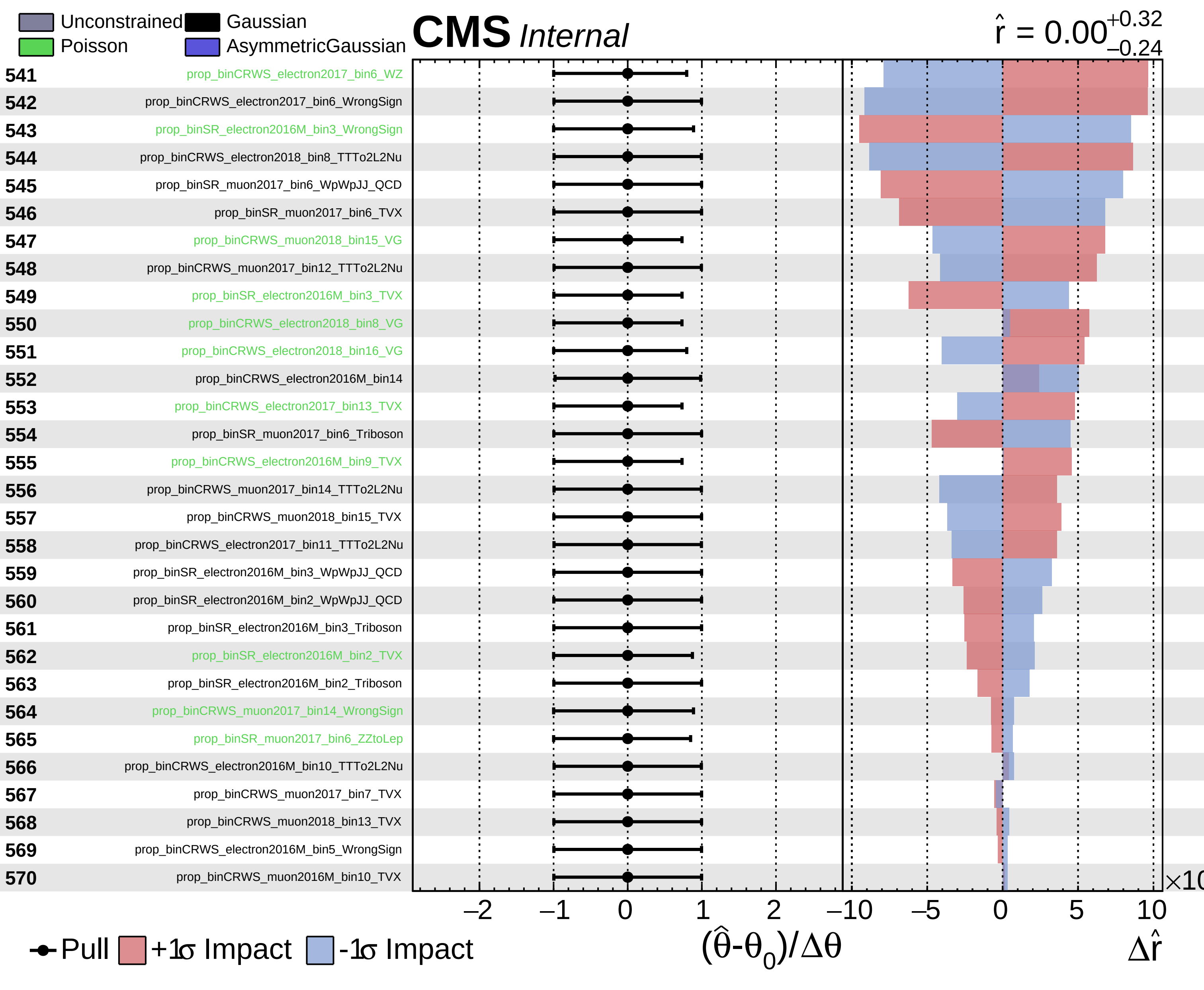
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.24
451 prop_binSR_electron2017_bin11_TVX
452 prop_binCRTT_electron2018_bin16
453 prop_binSR_electron2016M_bin16_WrongSign
454 prop_binCRTT_electron2016M_bin16
455 prop_binCRWS_muon2018_bin15_Fake
456 prop_binCRTT_muon2017_bin19
457 prop_binCRTT_muon2017_bin11
458 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
459 prop_binCRTT_electron2016M_bin12
460 prop_binCRTT_muon2016M_bin18
461 prop_binCRTT_electron2017_bin11
462 prop_binCRTT_muon2017_bin13
463 prop_binCRTT_muon2016M_bin14
464 prop_binCRTT_muon2016M_bin16
465 prop_binCRTT_electron2017_bin17
466 prop_binCRWS_muon2018_bin13_Fake
467 prop_binSR_electron2017_bin19_ZZtoLep
468 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
469 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
470 prop_binCRWS_muon2018_bin5_WrongSign
471 prop_binCRTT_electron2017_bin5
472 prop_binCRTT_muon2017_bin3
473 prop_binSR_muon2017_bin6_WrongSign
474 lep_2016M
475 prop_binCRWS_muon2017_bin13_WrongSign
476 prop_binSR_electron2017_bin14_ZZtoLep
477 prop_binCRWS_electron2017_bin14_WrongSign
478 prop_binSR_muon2017_bin6_TTTo2L2Nu
479 prop_binCRTT_electron2018_bin5
480 prop_binCRWS_electron2016M_bin16
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



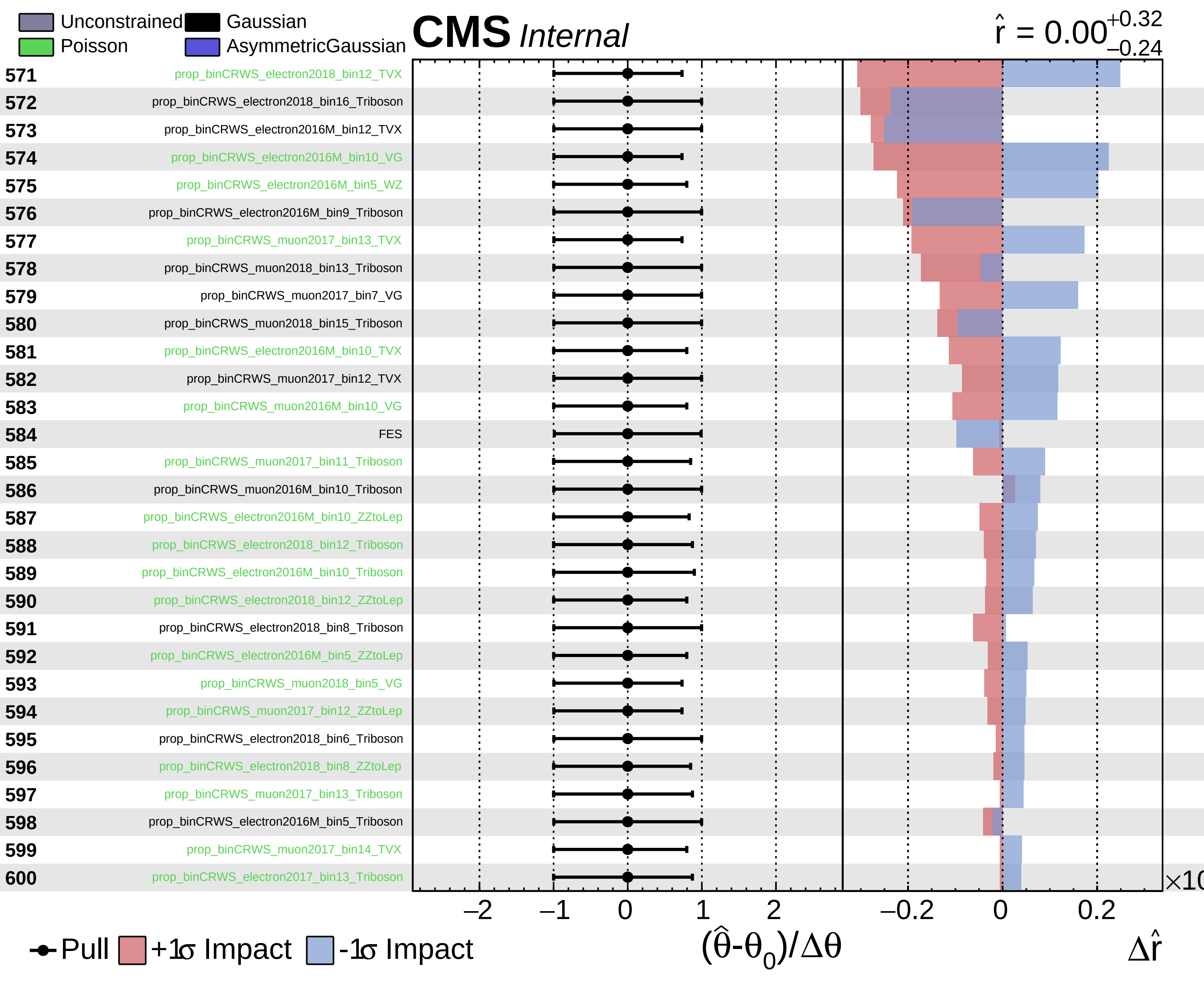
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
481 prop_binCRTT_muon2017_bin16
482 prop_binCRWS_electron2016M_bin11
483 prop_binSR_electron2016M_bin16_Triboson
484 prop_binCRTT_muon2017_bin9
485 prop_binSR_electron2016M_bin2_WZ
486 prop_binCRTT_electron2016M_bin18
487 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
488 prop_binCRWS_muon2017_bin16
489 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
490 prop_binCRWS_muon2017_bin19
491 prop_binCRTT_electron2016M_bin5
492 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
493 prop_binCRWS_electron2016M_bin12_Fake
494 prop_binCRWS_muon2016M_bin10_WrongSign
495 prop_binCRWS_muon2017_bin7_WrongSign
496 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
497 prop_binSR_electron2016M_bin3_VG
498 prop_binSR_electron2016M_bin2_WrongSign
499 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
500 lep_2017
501 prop_binSR_electron2016M_bin2_TTTo2L2Nu
502 prop_binSR_electron2016M_bin3_TTTo2L2Nu
503 prop_binCRWS_electron2016M_bin10_WrongSign
504 prop_binCRWS_muon2017_bin18
505 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
506 prop_binCRWS_muon2017_bin7_Fake
507 prop_binCRWS_electron2017_bin11_Fake
508 prop_binCRWS_electron2017_bin13_Fake
509 prop_binCRTT_muon2016M_bin6
510 prop_binCRTT_electron2017_bin12
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



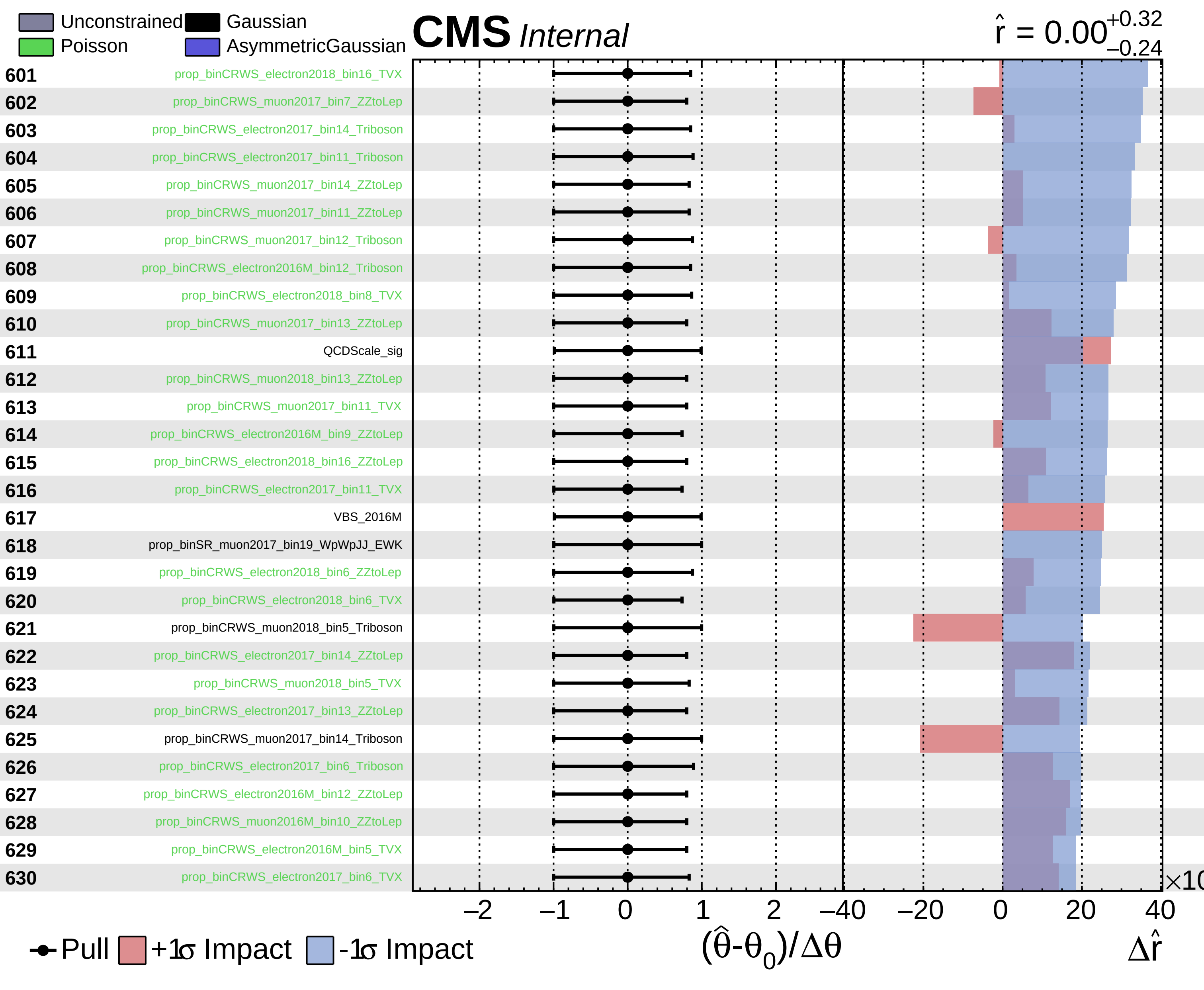
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
511 prop_binCRWS_muon2017_bin15
512 prop_binCRTT_electron2017_bin13
513 prop_binCRTT_electron2016M_bin9
514 prop_binCRWS_muon2018_bin13_WrongSign
515 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
516 prop_binCRWS_electron2016M_bin13
517 prop_binCRWS_electron2018_bin16_WrongSign
518 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
519 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
520 prop_binCRWS_electron2016M_bin18
521 prop_binCRWS_electron2016M_bin12_WrongSign
522 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
523 prop_binCRWS_electron2017_bin11_WrongSign
524 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
525 prop_binSR_electron2016M_bin3_ZZtoLep
526 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
527 prop_binCRWS_muon2017_bin12_Fake
528 prop_binCRWS_electron2018_bin6_WrongSign
529 prop_binCRWS_electron2017_bin13_WrongSign
530 prop_binCRWS_muon2018_bin15_WrongSign
531 prop_binSR_electron2016M_bin2_VG
532 prop_binCRWS_electron2016M_bin9_WrongSign
533 prop_binCRWS_electron2018_bin12_WrongSign
534 prop_binSR_electron2016M_bin2_ZZtoLep
535 prop_binCRWS_muon2017_bin12_WrongSign
536 prop_binCRWS_electron2018_bin16_Fake
537 prop_binCRWS_electron2018_bin8_WrongSign
538 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
539 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
540 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



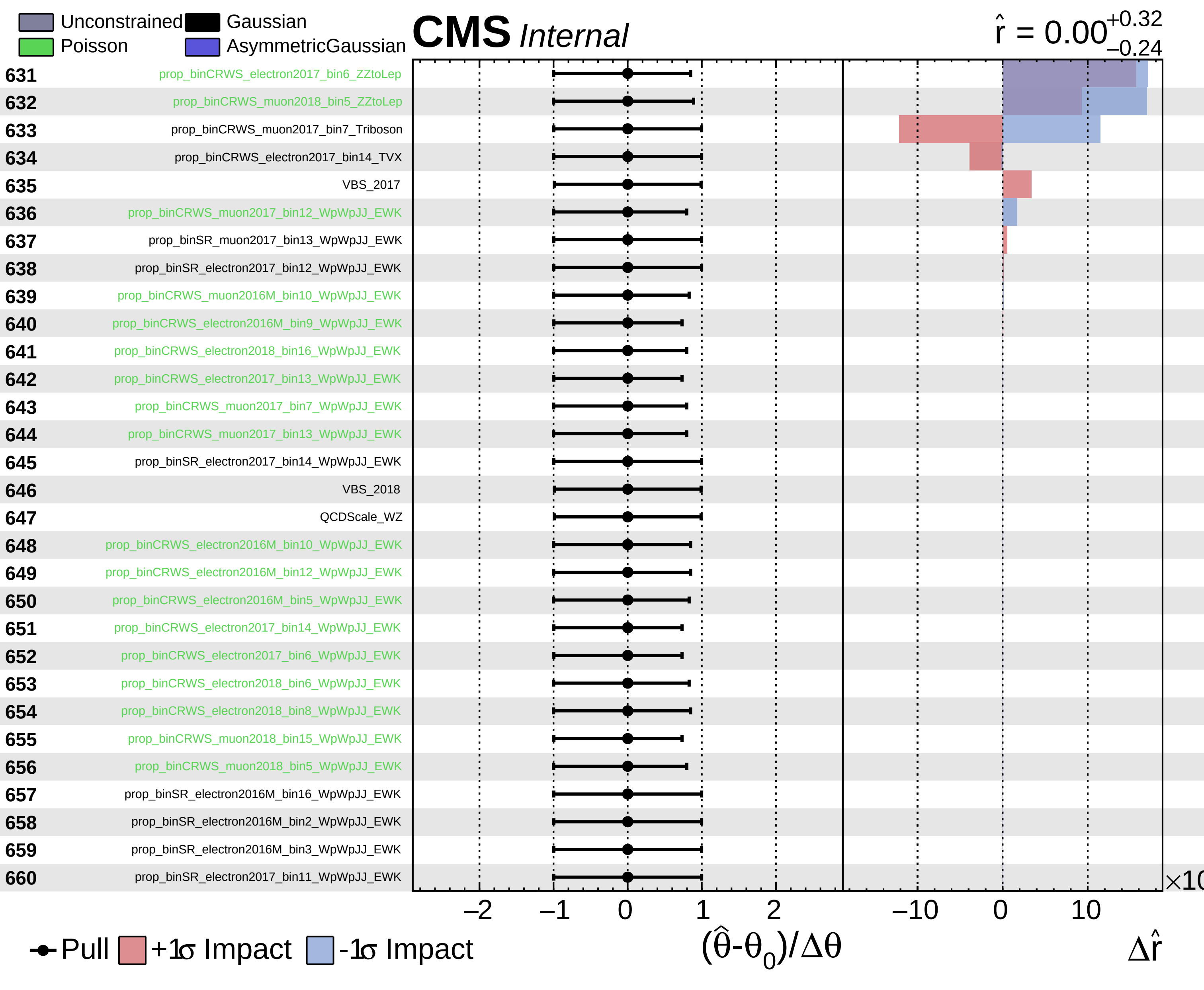
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
541 prop_binCRWS_electron2017_bin6_WZ
542 prop_binCRWS_electron2017_bin6_WrongSign
543 prop_binSR_electron2016M_bin3_WrongSign
544 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
545 prop_binSR_muon2017_bin6_WpWpJJ_QCD
546 prop_binSR_muon2017_bin6_TVX
547 prop_binCRWS_muon2018_bin15_VG
548 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
549 prop_binSR_electron2016M_bin3_TVX
550 prop_binCRWS_electron2018_bin8_VG
551 prop_binCRWS_electron2018_bin16_VG
552 prop_binCRWS_electron2016M_bin14
553 prop_binCRWS_electron2017_bin13_TVX
554 prop_binSR_muon2017_bin6_Triboson
555 prop_binCRWS_electron2016M_bin9_TVX
556 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
557 prop_binCRWS_muon2018_bin15_TVX
558 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
559 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
560 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
561 prop_binSR_electron2016M_bin3_Triboson
562 prop_binSR_electron2016M_bin2_TVX
563 prop_binSR_electron2016M_bin2_Triboson
564 prop_binCRWS_muon2017_bin14_WrongSign
565 prop_binSR_muon2017_bin6_ZZtoLep
566 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
567 prop_binCRWS_muon2017_bin7_TVX
568 prop_binCRWS_muon2018_bin13_TVX
569 prop_binCRWS_electron2016M_bin5_WrongSign
570 prop_binCRWS_muon2016M_bin10_TVX
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



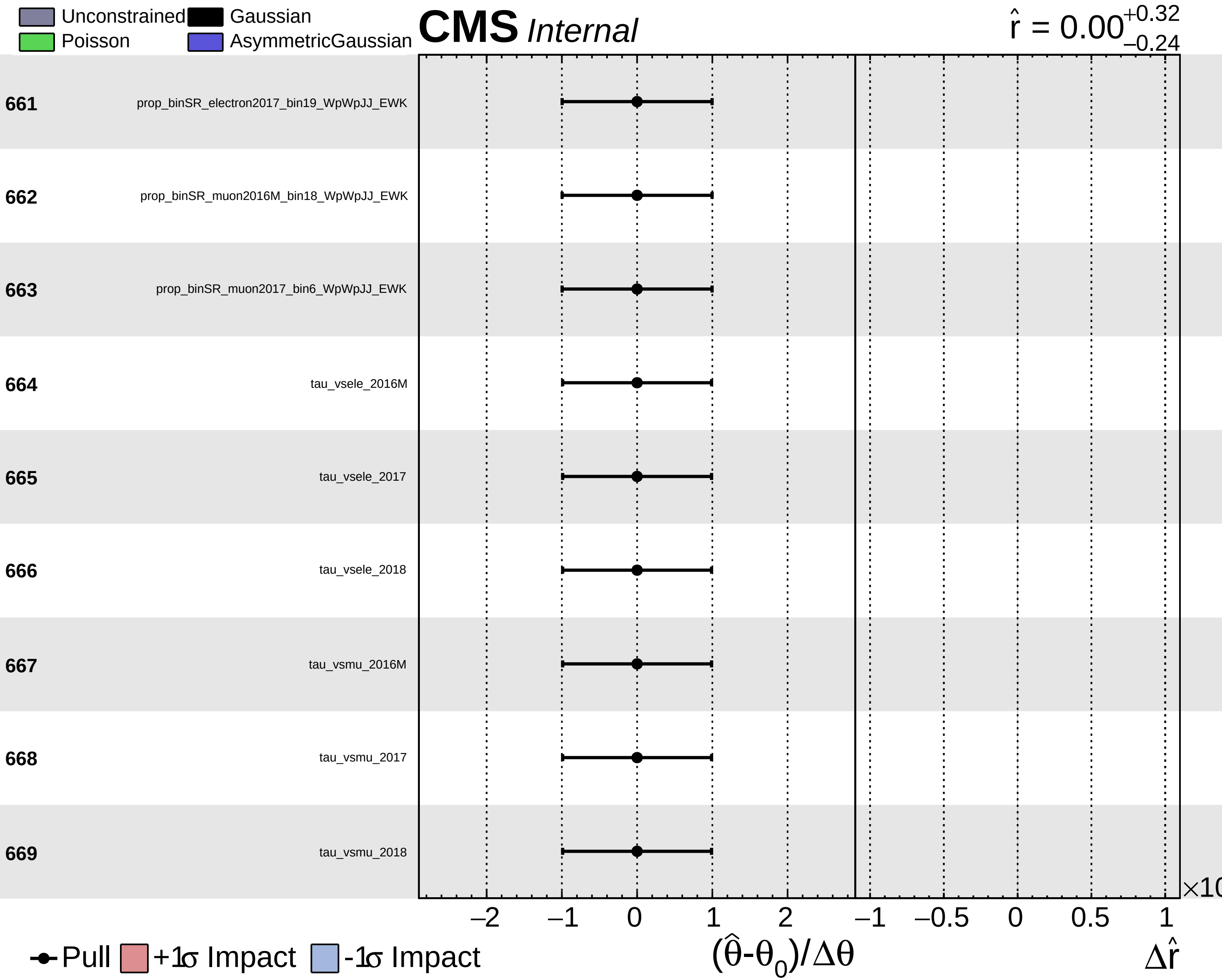
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
571 prop_binCRWS_electron2018_bin12_TVX
572 prop_binCRWS_electron2018_bin16_Triboson
573 prop_binCRWS_electron2016M_bin12_TVX
574 prop_binCRWS_electron2016M_bin10_VG
575 prop_binCRWS_electron2016M_bin5_WZ
576 prop_binCRWS_electron2016M_bin9_Triboson
577 prop_binCRWS_muon2017_bin13_TVX
578 prop_binCRWS_muon2018_bin13_Triboson
579 prop_binCRWS_muon2017_bin7_VG
580 prop_binCRWS_muon2018_bin15_Triboson
581 prop_binCRWS_electron2016M_bin10_TVX
582 prop_binCRWS_muon2017_bin12_TVX
583 prop_binCRWS_muon2016M_bin10_VG
584 FES
585 prop_binCRWS_muon2017_bin11_Triboson
586 prop_binCRWS_muon2016M_bin10_Triboson
587 prop_binCRWS_electron2016M_bin10_ZZtoLep
588 prop_binCRWS_electron2018_bin12_Triboson
589 prop_binCRWS_electron2016M_bin10_Triboson
590 prop_binCRWS_electron2018_bin12_ZZtoLep
591 prop_binCRWS_electron2018_bin8_Triboson
592 prop_binCRWS_electron2016M_bin5_ZZtoLep
593 prop_binCRWS_muon2018_bin5_VG
594 prop_binCRWS_muon2017_bin12_ZZtoLep
595 prop_binCRWS_electron2018_bin6_Triboson
596 prop_binCRWS_electron2018_bin8_ZZtoLep
597 prop_binCRWS_muon2017_bin13_Triboson
598 prop_binCRWS_electron2016M_bin5_Triboson
599 prop_binCRWS_muon2017_bin14_TVX
600 prop_binCRWS_electron2017_bin13_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.24
601 prop_binCRWS_electron2018_bin16_TVX
602 prop_binCRWS_muon2017_bin7_ZZtoLep
603 prop_binCRWS_electron2017_bin14_Triboson
604 prop_binCRWS_electron2017_bin11_Triboson
605 prop_binCRWS_muon2017_bin14_ZZtoLep
606 prop_binCRWS_muon2017_bin11_ZZtoLep
607 prop_binCRWS_muon2017_bin12_Triboson
608 prop_binCRWS_electron2016M_bin12_Triboson
609 prop_binCRWS_electron2018_bin8_TVX
610 prop_binCRWS_muon2017_bin13_ZZtoLep
611 QCDScale_sig
612 prop_binCRWS_muon2018_bin13_ZZtoLep
613 prop_binCRWS_muon2017_bin11_TVX
614 prop_binCRWS_electron2016M_bin9_ZZtoLep
615 prop_binCRWS_electron2018_bin16_ZZtoLep
616 prop_binCRWS_electron2017_bin11_TVX
617 VBS_2016M
618 prop_binSR_muon2017_bin19_WpWpJJ_EWK
619 prop_binCRWS_electron2018_bin6_ZZtoLep
620 prop_binCRWS_electron2018_bin6_TVX
621 prop_binCRWS_muon2018_bin5_Triboson
622 prop_binCRWS_electron2017_bin14_ZZtoLep
623 prop_binCRWS_muon2018_bin5_TVX
624 prop_binCRWS_electron2017_bin13_ZZtoLep
625 prop_binCRWS_muon2017_bin14_Triboson
626 prop_binCRWS_electron2017_bin6_Triboson
627 prop_binCRWS_electron2016M_bin12_ZZtoLep
628 prop_binCRWS_muon2016M_bin10_ZZtoLep
629 prop_binCRWS_electron2016M_bin5_TVX
630 prop_binCRWS_electron2017_bin6_TVX
−2 −1 0 1 2
−40 −20 0 20 40
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ₀)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
631 prop_binCRWS_electron2017_bin6_ZZtoLep
632 prop_binCRWS_muon2018_bin5_ZZtoLep
633 prop_binCRWS_muon2017_bin7_Triboson
634 prop_binCRWS_electron2017_bin14_TVX
635 VBS_2017
636 prop_binCRWS_muon2017_bin12_WpWpJJ_EWK
637 prop_binSR_muon2017_bin13_WpWpJJ_EWK
638 prop_binSR_electron2017_bin12_WpWpJJ_EWK
639 prop_binCRWS_muon2016M_bin10_WpWpJJ_EWK
640 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
641 prop_binCRWS_electron2018_bin16_WpWpJJ_EWK
642 prop_binCRWS_electron2017_bin13_WpWpJJ_EWK
643 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
644 prop_binCRWS_muon2017_bin13_WpWpJJ_EWK
645 prop_binSR_electron2017_bin14_WpWpJJ_EWK
646 VBS_2018
647 QCDScale_WZ
648 prop_binCRWS_electron2016M_bin10_WpWpJJ_EWK
649 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
650 prop_binCRWS_electron2016M_bin5_WpWpJJ_EWK
651 prop_binCRWS_electron2017_bin14_WpWpJJ_EWK
652 prop_binCRWS_electron2017_bin6_WpWpJJ_EWK
653 prop_binCRWS_electron2018_bin6_WpWpJJ_EWK
654 prop_binCRWS_electron2018_bin8_WpWpJJ_EWK
655 prop_binCRWS_muon2018_bin15_WpWpJJ_EWK
656 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
657 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
658 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
659 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
660 prop_binSR_electron2017_bin11_WpWpJJ_EWK
−2 −1 0 1 2
−10 0 10
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.24}$
661 prop_binSR_electron2017_bin19_WpWpJJ_EWK
662 prop_binSR_muon2016M_bin18_WpWpJJ_EWK
663 prop_binSR_muon2017_bin6_WpWpJJ_EWK
664 tau_vsele_2016M
665 tau_vsele_2017
666 tau_vsele_2018
667 tau_vsmu_2016M
668 tau_vsmu_2017
669 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$